

Study on the Status of Building Code Implementation in 2015 Earthquake Affected District



Report by:



Primarily funded by:



May 2021

About HRRP

The Housing Recovery and Reconstruction Platform (HRRP) was established in December 2015 to take over supporting coordination of the post-earthquake housing reconstruction from the Nepal Shelter Cluster, as it returned to the pre-earthquake format as a standard cluster. HRRP provides coordination support services for the National Reconstruction Authority (NRA), Building and Grant Management and Local Infrastructure (GMALI) Central Level Project Implementation Units (CLPIUs), other relevant government authorities, and Partner Organisations (POs). HRRP is currently in its fourth phase and will continue its coordination services until July 2021. HRRP is primarily funded by UKAid Nepal and led by CRS Nepal with technical leadership from NSET.

The HRRP works at municipal, district, sub-national, and national levels, providing support primarily to the 14 districts most affected by the earthquake, as well as roving support to the 18 districts moderately affected by the earthquake.

Areas of Focus

The HRRP's main areas of focus are:

- Monitoring and documenting the housing reconstruction process.
- Facilitating coordination and experience sharing to support improvements in coverage and quality of socio-technical assistance.
- Supporting collective planning and roll out of additional support for vulnerable households.
- Addressing gaps and duplications.
- Advocacy, communications, and research.



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Contents

Acknowledgement	ii
Executive Summary	iii
1. Introduction	1
1.1 Background	1
1.2 National Building Code	1
1.3 Evolving Municipal Structure	2
1.4 2015 Gorkha Earthquake	2
1.5 Objective	4
1.6 Limitations	4
2. Methodology	5
2.1 Quantitative Assessment	5
2.2 Selected Municipality and Sample Size	5
3. Data Exploration	6
3.1 Demographic profile	6
3.1.1 Gender	6
3.1.2 Profession	6
3.1.3 Municipality Declaration	6
3.1.4 Population density	6
3.2 Housing Statistic	7
3.2.1 Housing status	7
3.2.2 Housing Typology	7
3.3 Human Resource	9
3.3.1 Technical staff	9
3.3.2 Masons and petty contractors	10
3.3.3 Consultant Engineer	10
3.4 Findings	11
3.4.1 Technical Assistance	11
3.4.2 Retrofitting training	13
3.4.3 Status of Building Code Implementation	18
3.4.4 Building Permit Process	19
3.4.5 Municipal Action Plan	24
3.5 Analysis and interpretation	26
3.5.1 Number of municipal technical staff Vs Houses construction per year	26
3.5.2 Mason and petty contractor Vs Houses construction per year	26
3.5.3 Correlation between the number of municipal technical staffs and BCI	27
3.6 Identification of the most critical issues and Challenges	28
4. Conclusion	29
4.1 Recommendation and Way forward	29
5. Annexes	31
Annex 1 - Common building typology in Nepal (DUDBC, 2009)	31
Annex 2 - Existing National Building Code in Nepal, DUDBC	32
Annex 3 - Survey Questionnaire	34

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Executive Summary

This study presents the findings of the municipality level survey to understand the status of the Building Code Implementation across the districts affected by the 25 April 2015 Gorkha earthquake. Government of Nepal with financial assistance from the United Nations Development Programme and United Nations Centre for Human Settlement (UN-HABITAT) started working on Nepal national building code in 1993 which led to development of Building Code in 1994.

The Building Code Development Project involved seismic hazard mapping, risk assessment, legal framework, inclusion of alternative building materials and technologies, public awareness, and its dissemination. The Ministry of Physical Planning and Work, an authorized body for the implementation of the code published a notice in the Gazette in 2006, which made mandatory implementation of NBC in all municipalities and administrative headquarter of 28 Village Development Committees (VDCs) in Nepal. But to the irony, the 2015 Gorkha earthquake destroyed large number of private houses. The damage exposed the weaknesses of houses that did not have any seismic-resistant features or were not in accordance with the building codes. Thus, the need for the study on the status of building code implementation was required. The study was conducted with the objective of understanding the status of the building code implementation in the rural/urban municipalities and to assess awareness level regarding the National Building Code at the municipal level.

For the quantitative survey, 34 municipalities from 24 districts listed among the most affected to least affected were selected. The sample size included 18 rural municipalities and 16 urban municipalities.

Main findings from the study includes:

In trend of construction of masonry structure was found to be dominant in both urban and rural municipalities whereas the rate of construction of new houses was found to be more in the rural municipalities. As per the result of both RCC and masonry structure within the study area, it can be seen almost 20% of the structures present are not covered by the MRT (NBC 202, NBC 203, and NBC 205) and lies under the need of professionally designed structure. Considering the number of technical staffs on an average 9 and 12 engineers were available in rural and urban municipalities respectively.

Almost 61% of the municipalities had conducted training/awareness program on BCI to the engineers but no such programs were targeted to social mobilizers and consultant engineers and municipality staffs as 76% of the municipality have claimed to have little awareness on BCI among municipal staffs. There were very few organizations providing technical support on safer construction and the major activity included training to engineers and mason. Also, the technical training was mainly focused on new construction and less programs focused on retrofitting. Among the technical assistance activity conducted by municipality about 88% of the municipality had conducted door to door technical assistance followed by orientation to the houseowner on earthquake resistant construction. The need for the orientation to the community leader and local elected representative were observed.

Only 20 municipalities (approximately 59%) among the surveyed have implemented the building code but among these, majority of the municipalities focused on the semi-urban and urban areas only. out of 34 surveyed municipality, 18 municipalities had separate building

permit cell or unit to provide service to the residents. Similarly, 6 municipality had technical staffs dedicated to providing information or facilitate the building permit system. But 6 municipalities had no provision of assisting the building permit process. Only 11 (32%) out of 34 municipalities have plans to implement the building code into their building permit process.

Municipalities are planning on different awareness and trainings focused on building code and its importance to promote safer and resilient community.

1. Introduction

This study on the status of building code implementation in 2015 Gorkha Earthquake affected district is carried out by Housing Recovery and Reconstruction Platform (HRRP) to understand the building code implementation status across the earthquake affected districts.

1.1 Background

Nepal has a very diverse physiographic and climatic condition which influences the social economic status of the people including their housing style and structure. Low strength masonry such as stone in mud mortar, brick in mud mortar, and timber frame structures were and still are the most common building typologies in most of the rural area of Nepal. These types of houses were frequently constructed using the local materials and human resource. Similarly, in the urban area, low strength structures were most common mainly within core areas of Kathmandu valley followed by non-engineered reinforcement cement concrete frame structures. Annex 1 provides a description of each housing typology. In the absence of an appropriate implementation of building code, most buildings in Nepal were, and are still being, planned and constructed with little or no knowledge for seismic safety.

1.2 National Building Code

Overview

The 1988 earthquake in Nepal, and the resulting deaths and damage to both housing and schools, drew attention to the need for changes and improvement in the prevailing building design and construction methods was highly recognized. Accordingly, a Building Code Development Project was launched in 1992. The objectives of the project, among others, included the development of regulations and design documents for use by the planners and engineers in order to improve the seismic safety aspects of the existing buildings and to suggest safer building design and construction practices to be introduced/enforced gradually in Nepal. Government of Nepal with financial assistance from the United Nations Development Programme and United Nations Centre for Human Settlement (UN-HABITAT) started working on Nepal national building code in 1993 which led to development of Building Code in 1994.

The Building Code Development Project involved seismic hazard mapping, risk assessment, legal framework, inclusion of alternative building materials and technologies, public awareness, and its dissemination procedures. The nature and characteristics of the National Building Code (NBC) was developed considering the prevailing building typology, knowledge and skill of human resources, municipal capacity for seismic safety. The NBC has four distinct levels of applications. They are as follows:

- Level I: International state-of-art
- Level II: Professionally engineered structures
- Level III: Buildings of restricted size designed to simple rules-of-thumb
- Level IV: Remote rural buildings where control is impractical.

Initially, the 20 volumes draft codes with 4 levels of application were prepared in 1994. After the addition of 3 volumes of code by DUDBC a set of 23 volumes of NBC were approved by the GoN in 2003. The codes are varied in ways of difference in requirements and specifications. It ranges from NBC 000:1994 to NBC 114 and NBC 201 to 208 former and revised versions

as indicated Annex 2. The building code was reinforced through the Building Act, 1998 (first amendment 2007) and Building Regulation, 2008. The explanatory documents are such that an experienced overseer will be able to understand them and present sufficient details at the time of permit application to prove to a skilled appraiser at the Local Authority that the requirements have been met. The requirements are in terms of limits on spans and heights, minimum reinforcing, and member sizes, positioning of earthquake-resisting elements and other such rules.

The Ministry of Physical Planning and Work, an authorized body for the implementation of the code published a notice in the Gazette in 2006 which made the implementation of NBC mandatory in all municipalities and some Village Development Committees (VDCs) in Nepal. In 2002, prior to the formal entry into force of the code, Lalitpur Sub-Metropolitan City (now Lalitpur Metropolitan City) was the first municipality to implement NBC in 2001, followed by Kathmandu Metropolitan City in 2006 and Dharan Municipality in 2007 and few other municipalities out of 58 municipalities at the time. Though in later years, municipal structure in terms of geographical area and numbers kept increasing but the implementation was still not satisfactory across the municipalities.

1.3 Evolving Municipal Structure

The municipal structure in Nepal has changed many times in the past decades but municipality has not always meant urban as the indicator used to define a municipality has been primarily population. As a result, many municipalities remain partially, or even fully, rural in character.

In 1959, Nepal had just 10 municipalities. By 1992, this had increased to 36 municipalities and then further increased to 58 municipalities by 1996. In May 2014, with declaration of 14 additional municipalities it reached to 72 municipalities. Further in December 2014, 61 additional municipalities were declared. In the situation where building code implementation was limited to few municipalities, Nepal was struck by devastating earthquake on 25th April 2015 posing more challenge to the safer construction practice as it exposed the building construction practice in terms of seismic safety. Further, in September 2015, additional 26 municipalities were declared making it 159 municipalities in total. Under the new federal administration restructuring in Nepal the number of municipalities again changed in September 2017 with addition of 134 more municipalities, totaling to 293 municipalities. The numbers and type of municipalities were re-categorized into rural and urban. The urban municipalities were further classified as Metropolitan city, Sub-metropolitan City and Municipality. At present, Nepal has 6 metropolitan cities, 11 sub-metropolitan cities, 276 municipalities, and 460 rural municipalities. Since the introduction of the new administrative structure, all local bodies are now required to implement building code and issue building permits. However, situation has not changed much in the implementation of building code.

1.4 2015 Gorkha Earthquake

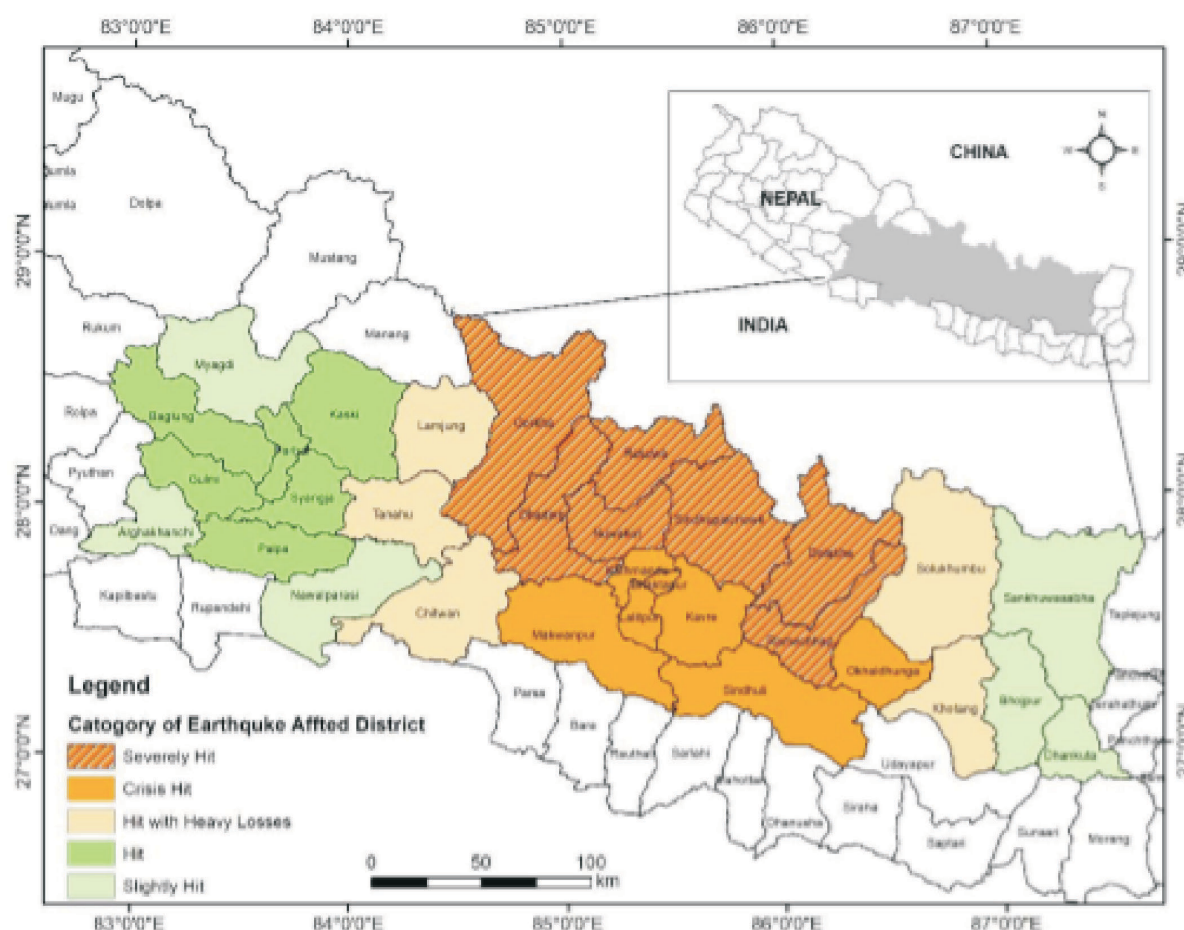
On April 25, 2015, a 7.8 magnitude earthquake struck Nepal. Following a strong shake on May 12 (7.3 magnitude), and a sequence of aftershocks, the Government of Nepal (GoN) reported a death toll of 8,700. The Post Disaster Needs Assessment (PDNA)¹ (http://www.worldbank.org/content/dam/Worldbank/document/SAR/nepal/PDNA_Volume_A_Final.pdf) conducted by the National Planning Commission identified 32 districts as earthquake affected. They

¹ Post Disaster Need Assessment, Vol 1, 2015

grouped the 32 affected districts into five major categories (NRA, 2015), seven districts being considered severely hit, seven being crisis hit, five being hit with heavy losses, six being hit, and the seven remaining districts being slightly affected (Figure 1).

- **Severely Hit:** Gorkha, Dhading, Rasuwa, Nuwakot, Sindhupalchok, Dolakha, and Ramechhap
- **Crisis Hit:** Kathmandu, Bhaktapur, Lalitpur, Kavre, Okhaldhunga, Sindhuli, and Makwanpur
- **Hit with Heavy Losses:** Lamjung, Tanahun, Chitwan, Solukhumbu, and Khotang
- **Hit:** Kaski, Parbat, Syangja, Palpa, Gulmi, and Baglung
- **Slightly Affected:** Myagdi, Arghakhanchi, Nawalpur, Parasi, Bhojpur, Dhankhuta, and Sankhuwasabha

Hundreds of historical and cultural monuments were either destroyed or extensively damaged and over half a million houses were destroyed. According to the Post Disaster Needs Assessment, about 95% of the houses damaged by the earthquake were of low-strength masonry. A large number of the houses damaged by the earthquake were traditional load bearing stone/brick in mud mortar structures in core cities followed by load bearing stone/brick in cement mortar structures and reinforced cement concrete (RCC) framed buildings



2015 Gorkha earthquake affected area based on severity of damage

especially in urban areas. The damage was mainly either due to neglected repairs or poor construction material and practices. The construction of multi-storied buildings using stilts on hilltops led to damage in a number of market towns. The damage exposed the weaknesses of houses that did not have any seismic-resistant features or were not in accordance with the building codes.

The National Reconstruction Authority) NRA was established in December 2015 to lead and facilitate the overall reconstruction including private housing with ‘Owner Driven” and ‘Build Back Better” approach. NRA with assistance and collaboration with different national and international stakeholder and organization developed technical guidance to ensure seismic safety and resilient construction. However, integration of these technical guidelines with regular building permit system has not be ensured. Under the medium to long term priorities for the Disaster Risk Reduction and Build Back Better, Government of has planned for development of a seismic policy and set up a network for seismic monitoring throughout the country and promote seismological research. This policy will include revision of building codes, development of building by-laws for all municipal areas, application of Mandatory Rule of Thumb (MRT) in rural areas, and development of risk-sensitive land use plans for all the municipalities of the country. Hence, the need to understand the present scenario of building code implementation in Nepal was identified.

1.5 Objective

- To understand the status of the building code implementation in the rural/urban municipalities.
- To comprehend the municipalities level of awareness regarding the National Building Code.
- To identify the existing human resources and their capacity for the implementation of the National Building Code in municipalities.
- Identification of municipalities within the study area for the institutionalization support on National Building Code.

1.6 Limitations

- Due to the increasing Covid-19 transmission risks and limited access to mobility the number of samples was decreased.

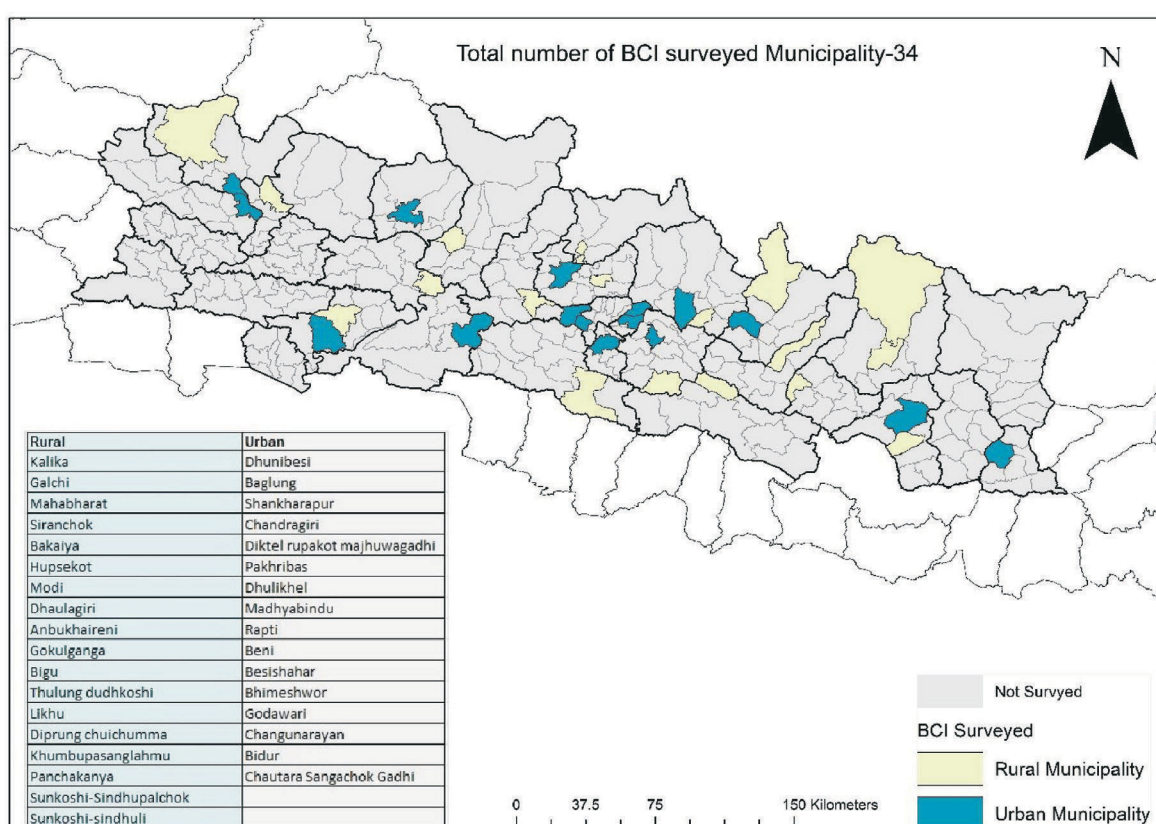
2. Methodology

2.1 Quantitative Assessment

The quantitative assessment questionnaire (Annex 3) was developed by the HRRP team. The questionnaire broadly covered the categories of the building code implementation status, existing building permit system, building construction rate, housing characteristic, available human resource including administrative, technical staffs, mason, petty contractor within the municipality periphery. Quantitative data was collected by HRRP district coordination team in June 2020.

2.2 Selected Municipality and Sample Size

A simple random sampling method was applied for selecting the sample from 32 earthquake affected districts. Out of total 282 municipalities, 34 municipalities were selected for the study. The sample size included 18 rural municipalities and 16 urban municipalities. Microsoft Excel was used for the analysis of the collected data.

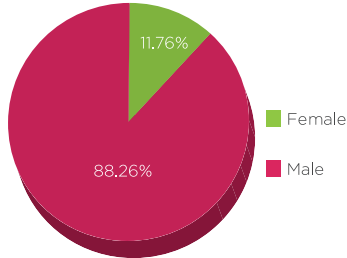


3. Data Exploration

3.1 Demographic profile

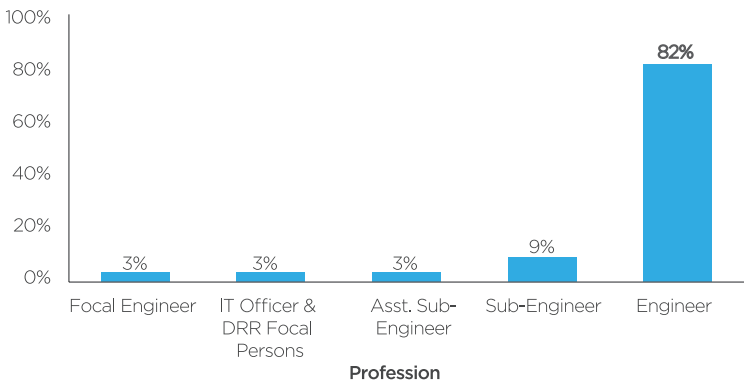
3.1.1 Gender

Among the total respondents of the survey 88.2% of the respondents were male and only 11.8% were female. Gender disaggregated data of respondents indicates that the male still outnumbered female in technical professions like engineering, IT etc.



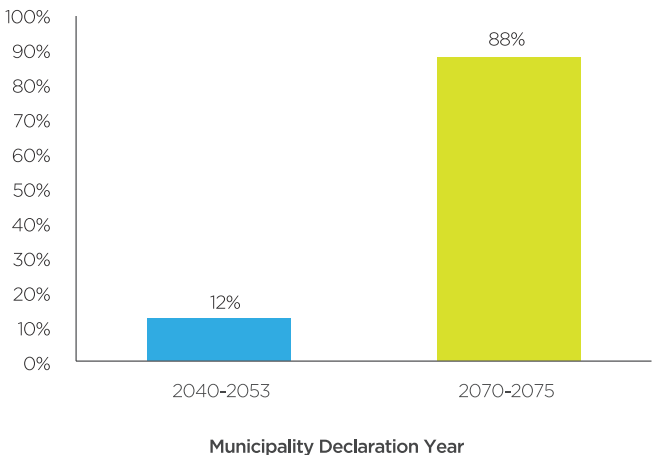
3.1.2 Profession

The majority of respondents of the survey were Municipal engineers who covered 82% of the total followed by Sub-engineer, Assistant sub-Engineer, IT officer & DRR focal person and focal engineer.



3.1.3 Municipality Declaration

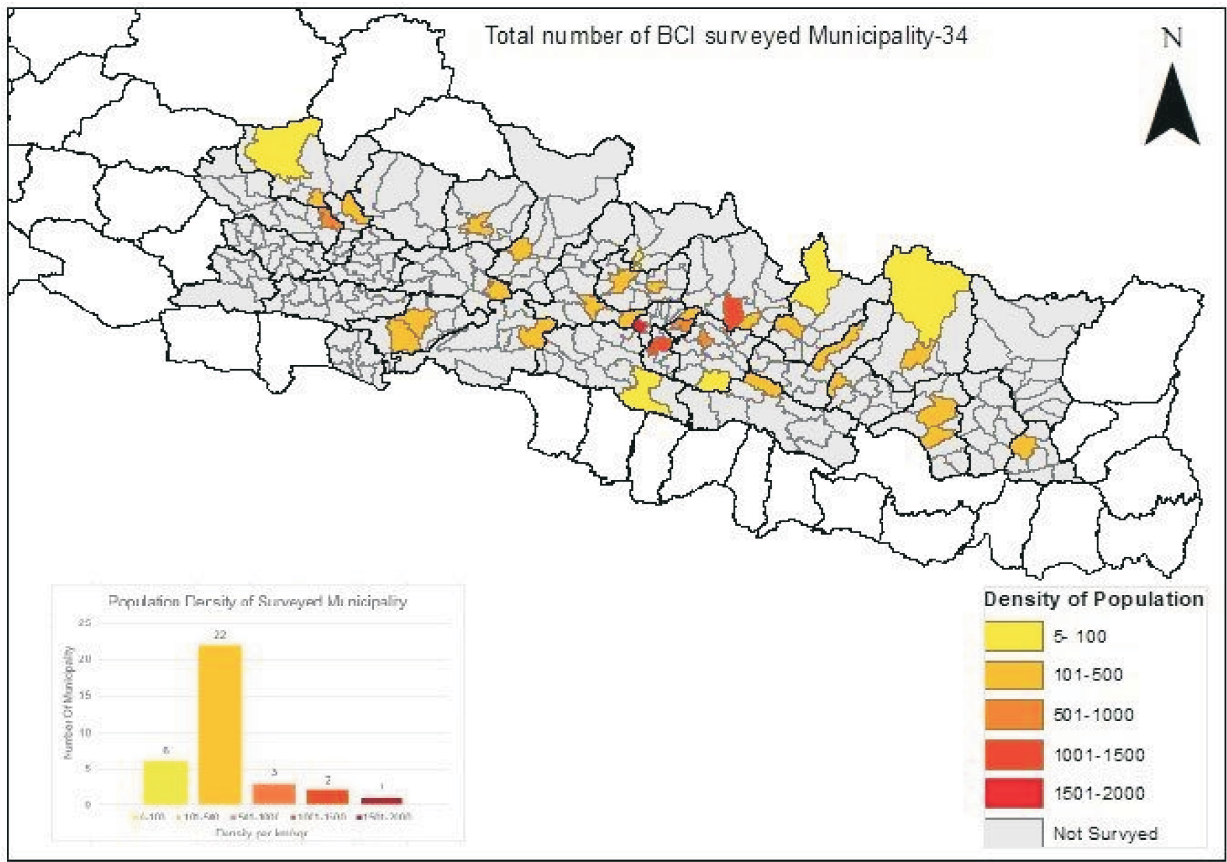
Of the 34 municipalities surveyed, 88% of local government were declared as Municipality / Rural Municipality in between 2070 BS to 2075 BS, while 12% were declared in between 2040 BS to 2053 BS. Under the new federal administration restructuring in Nepal the number of municipalities changed in 2074 B.S (September 2017) resulting in merging of the Village Development Committees(VDCs) to form rural/urban municipalities.



3.1.4 Population density

The map represents the population density of the surveyed municipalities. Here the black color represents the municipalities with least population density ranging from 0 – 250, whereas red color indicates the highest population density above 1000. Among the survey municipalities,

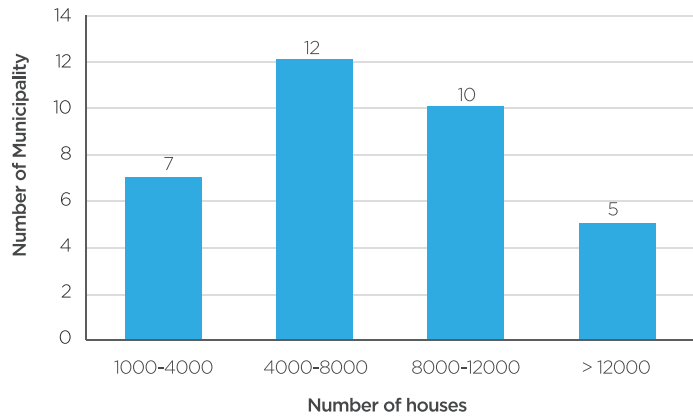
19 municipalities had population density ranging from 501-750 whereas 3 municipalities had population density higher than 1000.



3.2 Housing Statistic

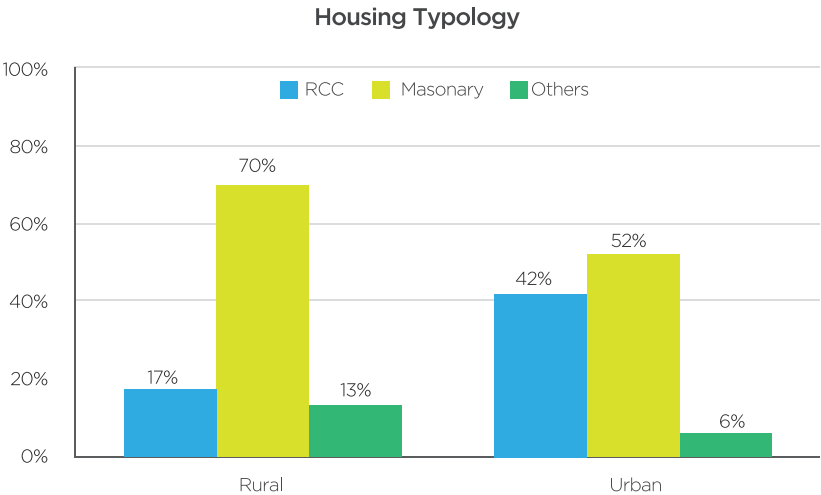
3.2.1 Housing status

Out of the 34 municipalities surveyed, it was found that 12 of them have a total number of houses between 4000 to 8000 while 10 municipalities have a number of houses from 8000 to 12000. Similarly, 5 municipalities reported a total number of houses more than 12000 whereas 7 of them only have a number of houses in between 1000 to 4000.



3.2.2 Housing Typology

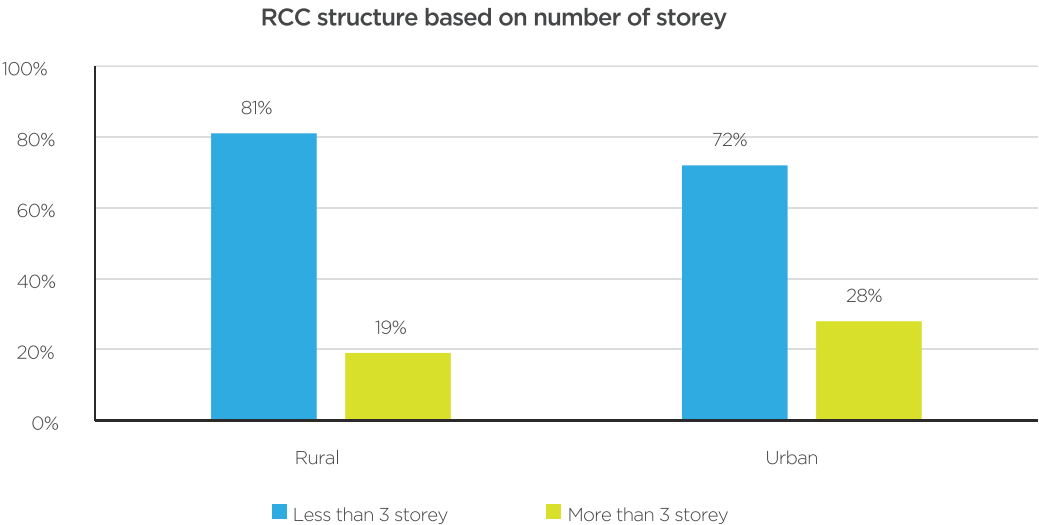
To understand the type of housing construction pattern in the study area, the housing typology was studied in terms of urban and rural area (hereafter, urban area refers to the urban municipalities and rural area refers to rural municipalities). As indicated in the figure, the rural municipality has dominance of masonry structure accounting of total 70% whereas in the urban municipalities the Masonry and RCC structure were found in similar ratio accounting 52% and 42% respectively. Here, the other structure indicates houses constructed using alternative materials such timber, hybrid structure etc.



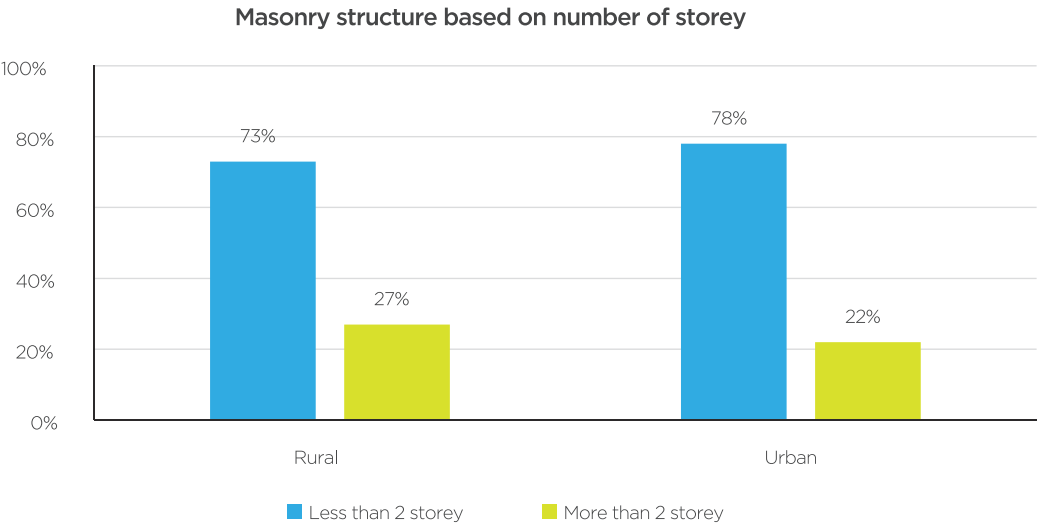
To further analyze the structure based on the number of storeys, the RCC and Masonry houses within the municipalities were categorized into:

- RCC (less than 3 and more than 3 storey)
- Masonry (less than 2 and more than 2 storey)

Among the existing houses within the surveyed municipalities, the trend of constructing RCC structure of less than 3 storeys was found higher in both rural area (81%) and urban area (72%). Similarly, the trend of constructing more than 3 storey RCC structure was found to be more in the urban municipalities.



Similarly, in case of masonry structure, houses with less than 2 storeys were found to be more in both rural (73%) and urban area (78%).

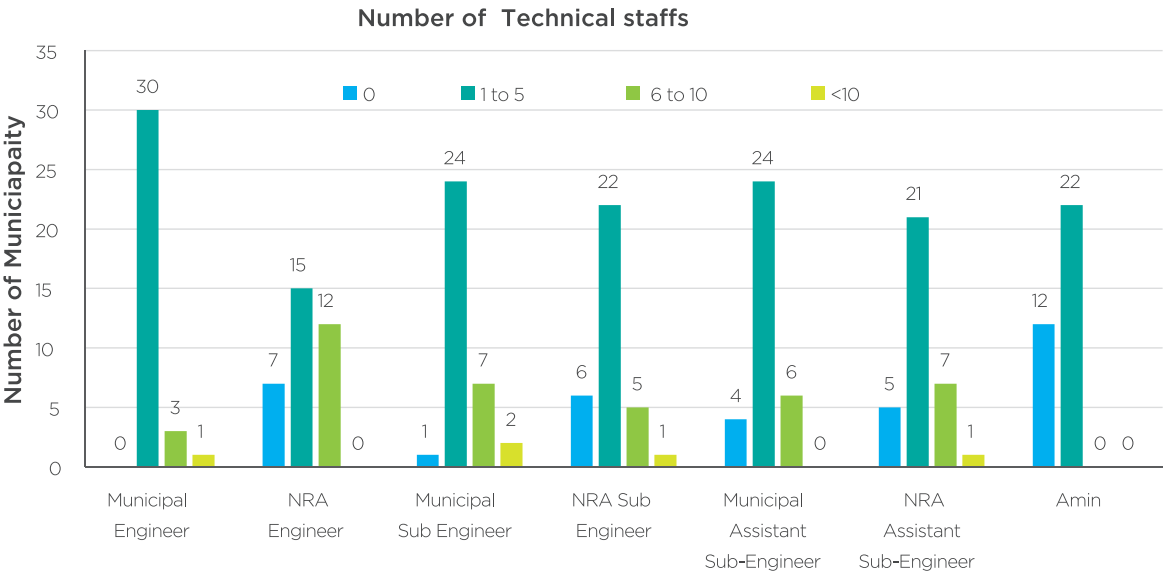


As per the result of both RCC and masonry structure within the study area, it can be seen almost 20% of the structures present are not covered by the MRT (NBC 202, NBC 203, and NBC 205) and lies under the need of professionally designed structure. Overall, the dominance of load bearing masonry structure was found in the surveyed municipalities with trend of RC structure rapidly increasing in urban areas with need of professionally designed structure to ensure safer construction practice.

3.3 Human Resource

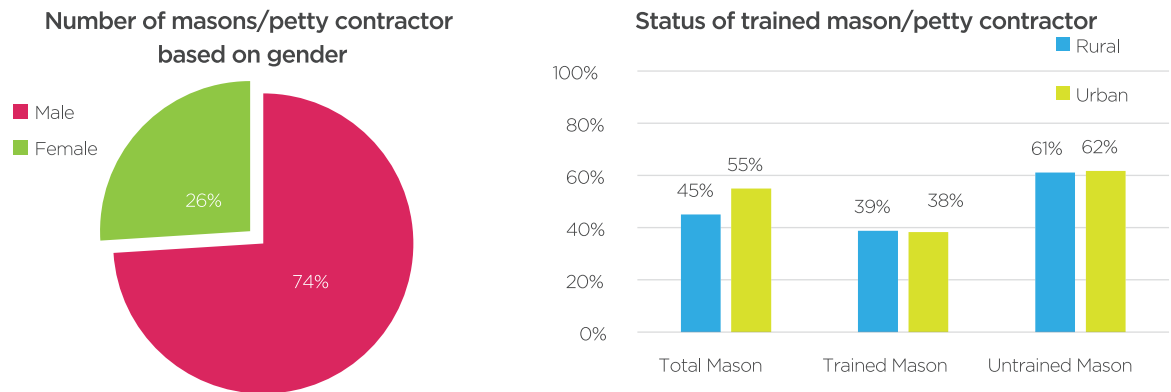
3.3.1 Technical staff

Regarding the available technical staff in the municipality, it is clearly seen from the graph majority of the staff presence ranges between 1-5. It also shows very few municipalities had presence of technical staff above 10. Following the recovery and reconstruction process in Nepal, the presence of technical staffs deployed from NRA has been significant in the surveyed municipalities. The presence of municipal technical staffs ranging between 1- 5 also might indicate that one technical staff is responsible for technical support in different sectors such as road, irrigation, building, water supply etc.



3.3.2 Masons and petty contractors

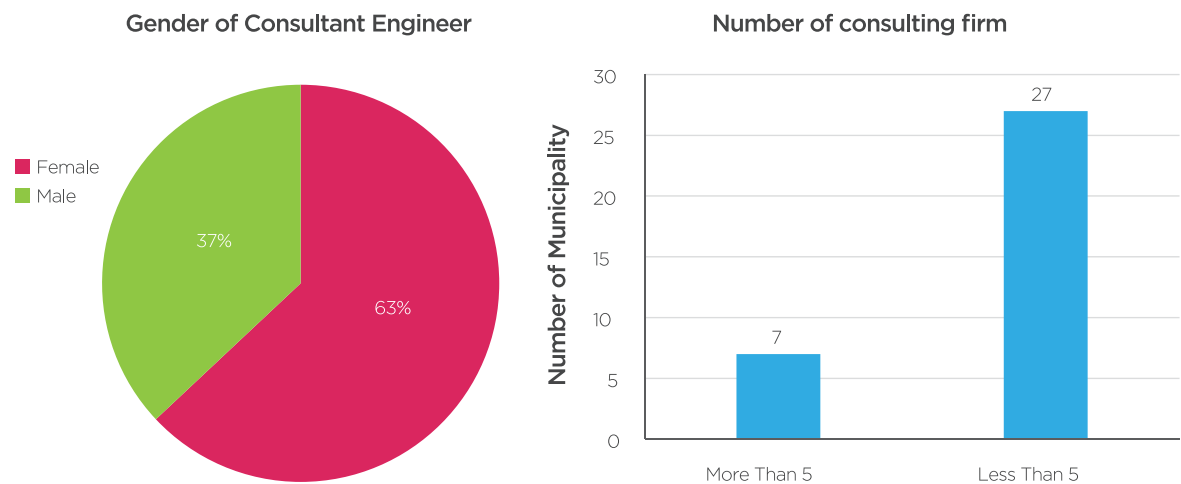
As per the finding from the survey, 74% were male and 26% represented female masons/petty contractor. Despite the high presence of male mason and petty contractors, the data also shows the gradual increase of female population construction work which was mainly considered male dominated profession.



In the basis of rural and urban areas, 55% of the mason/petty contractor were present in urban municipalities and 45% in rural municipality. Among these, 38% and 39% of the mason had received training in urban and rural municipalities respectively. Following the ongoing reconstruction work after the 2015 Gorkha earthquake, the number of mason/petty contractor receiving training in the rural area have increased in course of time but compared the trained mason the number of untrained masons is still significantly high.

3.3.3 Consultant Engineer

Regarding the consulting firms present in the surveyed municipalities, rural municipalities have total of 19 consulting firm with presence of 55 consulting engineers whereas in urban municipalities 129 consulting firms were found with presence of 460 consulting engineers. Among the total consultant engineers 63% were male and 37% were female consultant. This result also indicates the gradual increase of female consultant in the technical sector. It was also found that out of 34 surveyed municipalities, 7 municipalities had presence of more than 5 consulting firms whereas 27 municipalities had less than 5.

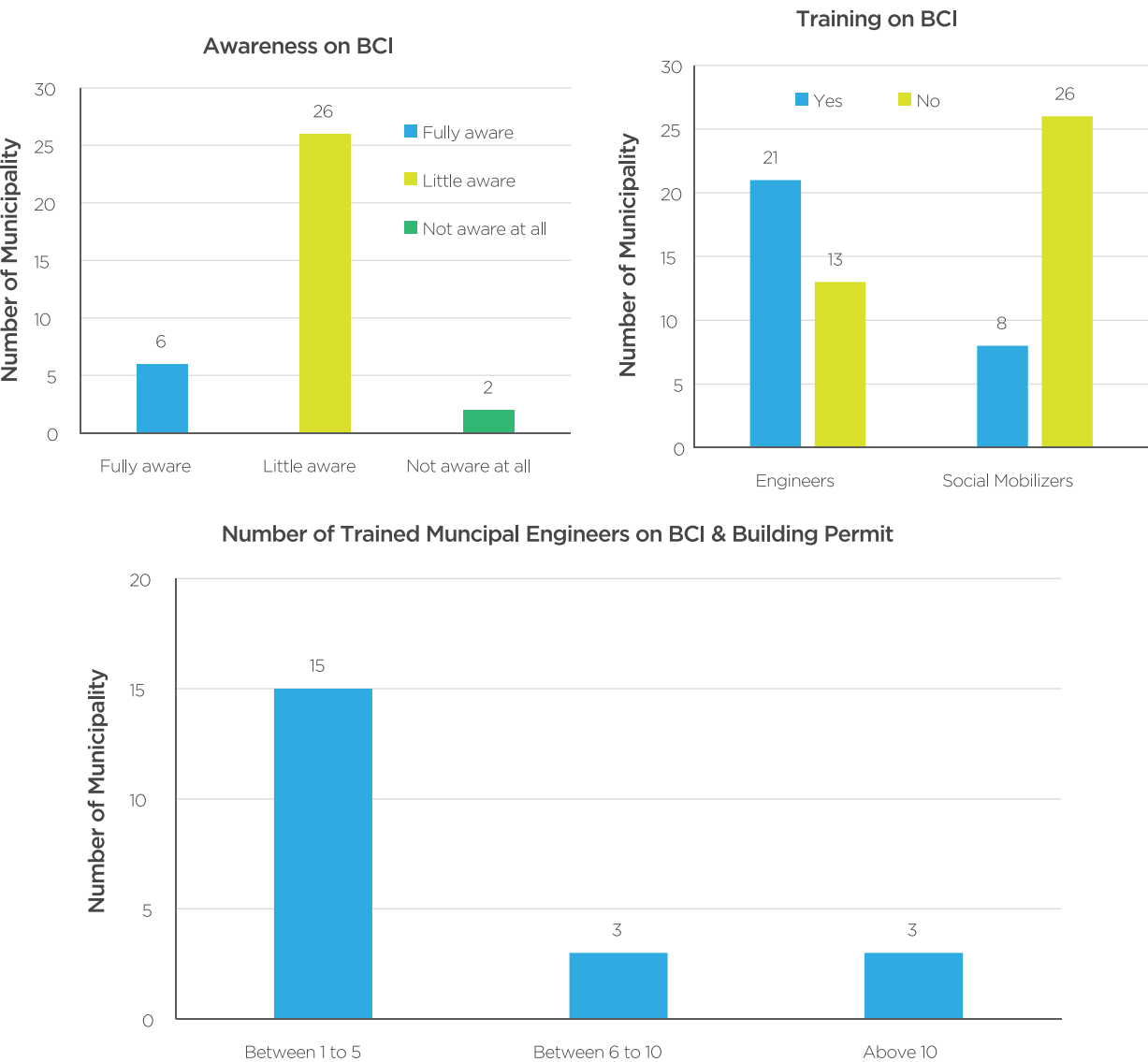


3.4 Findings

3.4.1 Technical Assistance

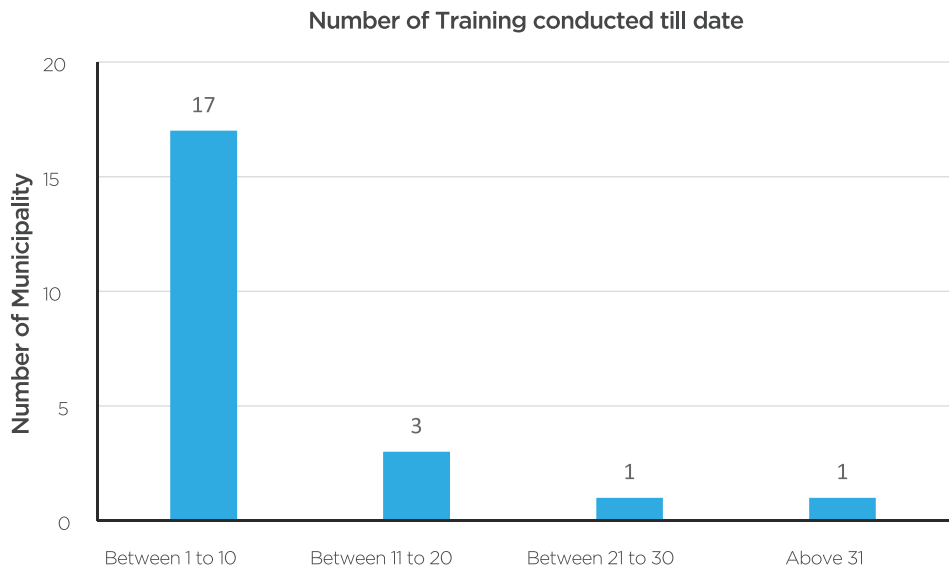
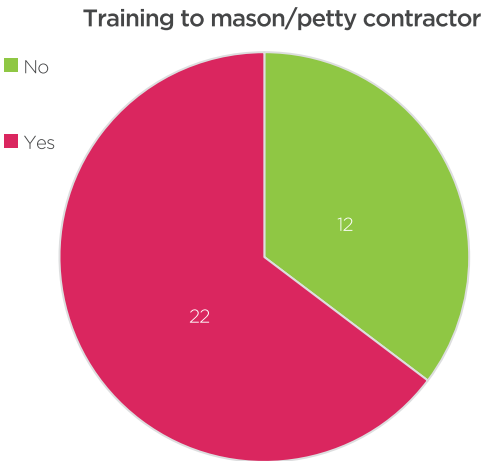
Training on BCI

Regarding the knowledge on BCI and Building permit system among the municipality staffs, we further asked question to categorize the level of awareness for which staffs of 26 municipalities excluding engineers/social mobilizers were found to be little aware regarding the building code implementation and building permit system. With more than 80% of the municipality with staff having least or no knowledge on BCI indicates the lack of importance of the building code which in turn will increase the risk of unsafe construction practice in the municipality. This also further indicates the needs of orientation and awareness program on BCI and its importance to all the municipalities staffs. Out of 34 municipalities surveyed, engineers of 21 municipalities have received training on building code implementation and building permit process whereas only 18 social mobilizers from 8 Municipalities had received training on building code implementation and building permit system. Among the 21 municipalities having conducted training for engineer, 15 municipalities have conducted training 1 to 5 times whereas 3 municipalities have conducted 6 to 10 times and above 10 times. The result shows that very few trainings are conducted for the technical staffs who are directly involved in the building permit process and responsible to ensure the safer construction practices which can in turn affect the quality of the construction activities within the municipalities



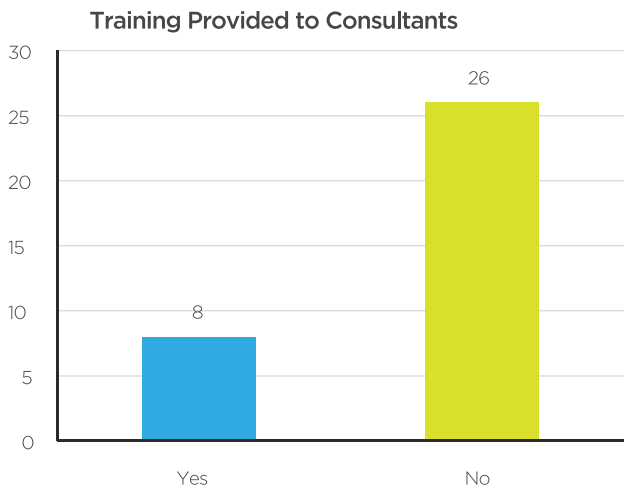
Mason/Petty Contractor Training

Training to mason and petty contractor has been conducted over past years by many governmental and non-governmental organizations. Among the study area, 22 municipalities confirmed mason training being conducted in their area whereas still 12 municipalities have no records of training the mason available in their area. In terms of gender, it was found a total of 4261 male masons and 1040 female masons have received training. According to the survey, most of the training was organized by NRA i.e., 3 whereas the rest of the organizations organized single training. Organizations which organized single training were Alliance Nepal, DUDBC, EHRP(JICA), Department of Cottage and Small Industries, Helvetas, JICA , Save the children, NRA, NSET, Caritas and Red cross. Majority of the municipality had conducted training ranging between 1 to 10.



Consultant/Designer Engineering Training

Regarding training on Earthquake resistant building design and construction has been provided at different time period. It was found that only 8 municipalities among the surveyed municipality have organized training focusing on the safer and resistant building design. Among the 8 municipalities, 7 of them have conducted only one training whereas remaining 1 municipality was found to have conducted such training 9 times. According to the survey, most of the training was organized by NRA i.e., 3 whereas organizations such as NSET, Build Change and Sahayata has organized one to two

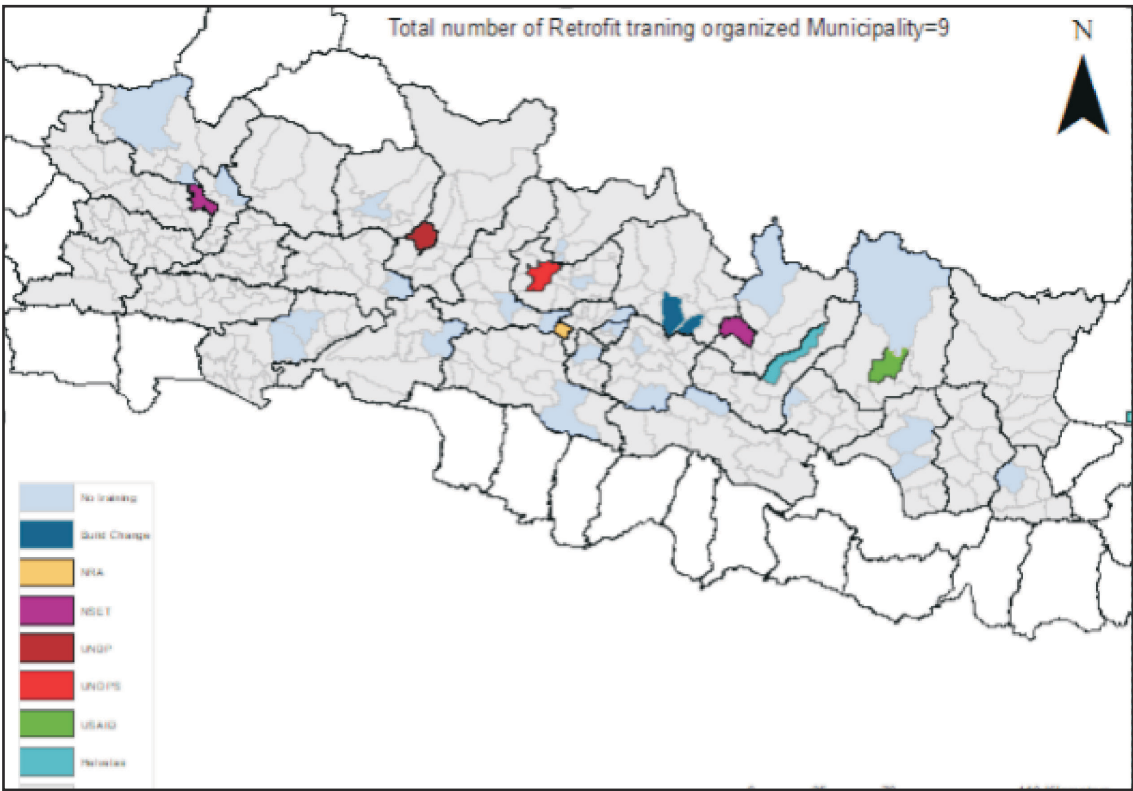
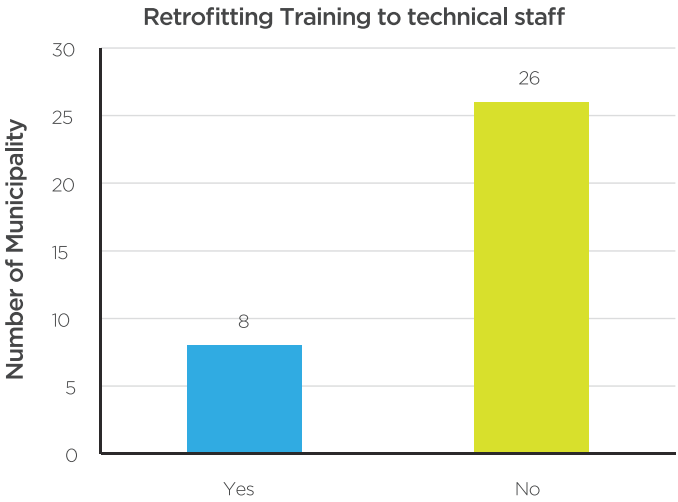


training. This in turn indicates that least priority given to the earthquake resistant training in municipalities and presence of few organization that provided such training at local level.

3.4.2 Retrofitting training

Technical staffs

In terms of providing retrofitting training to Engineers, only 8 municipalities (approximately 26.5%) reported their engineers have been trained in retrofitting while 26 municipalities (approximately 73.5%) reported that no retrofitting training have been conducted in their area. The majority of the trained engineers were male 55 and 12 female in 8 municipalities. The retrofitting training to engineers was organized by a number of organizations as indicated in the figure below. Till date organization such as NRA, NSET, Helvetas, EOI/ UNOPS, Build Change, UNDP, USAID has conducted the training as indicated in the coverage map below. To the irony, not a single municipality/rural municipality had taken initiation in organized retrofitting training in their respective area.

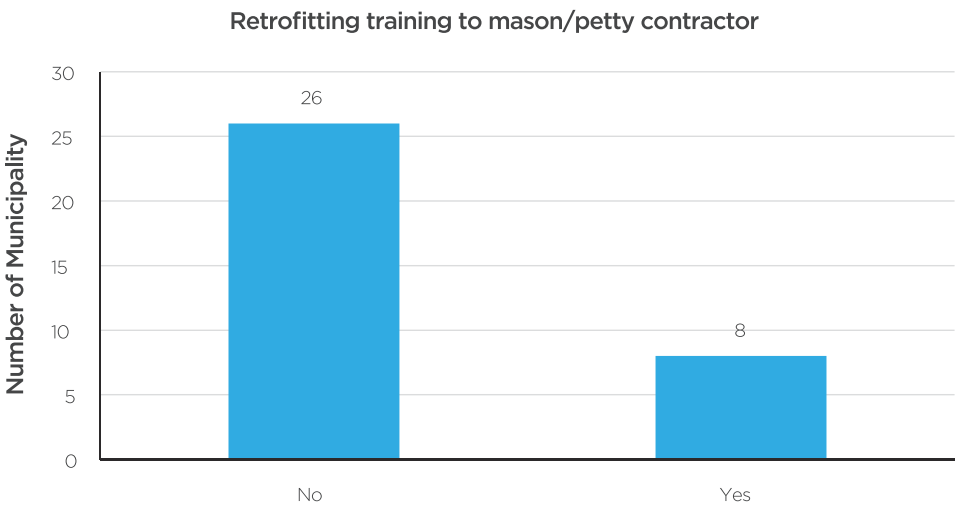


Coverage map of retrofitting training conducted by different organization

Retrofitting is a relatively new concept in Nepal and had been only used for the public buildings and schools. After the 2015 Gorkha earthquake, the concept and use of retrofitting private housing has immersed as one of the best alternatives to constructing a new house.

But unfortunately, there is low awareness and trust in the concept of retrofitting. The result also indicates the need of retrofitting training in Nepal as large proportion, close to three-quarters of the municipalities surveyed are yet to conduct training.**Masons/petty Contractor**

When asked about the retrofitting training to mason/petty contractors, it was found that only 8 municipalities (23.5%) were found to have conducted training on retrofitting to masons/petty contractor while 26 municipalities (76.5%) had no record of conducting such training. A large proportion, three-quarters of the municipalities surveyed still have to conduct training. Further, out of the 8 municipalities where retrofitting training have been conducted it was found that 7 municipalities conducted 1-5 times while only 1 municipality had conducted training more than 5 times. Among the trained masons/petty contractor majority were male occupying 77% and female only 23%. Overall, the result indicates the very less training on retrofitting to mason/petty contractor which can impact the retrofitting work in future.



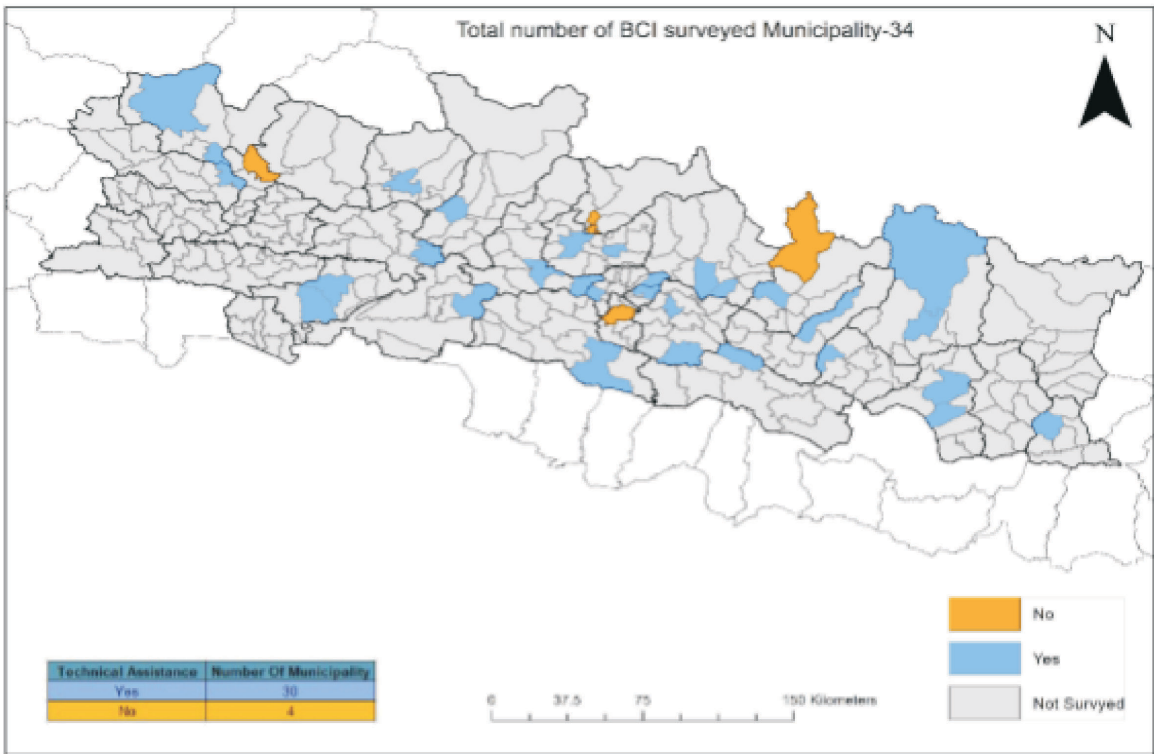
3.4.2.1 Technical Assistance

To understand the technical assistance provided at different level from household to the local level several questions were designed to cover the various modality used to increase the awareness on safer and resilient construction to disaster risk management. The technical assistance coverage map indicates that among the total surveyed municipalities, 30 municipalities have conducted at least one activity related to the technical assistance within the municipality. The specific technical assistance activity is further discussed individually.

a. Houseowner orientation on earthquake resistant design

To understand the level of awareness program at household level on the earthquake resistant designed and its importance, respondents were asked if any orientation programs on earthquake resistant design have been organized targeting such population. It was found that 21 surveyed municipalities have provided orientation on safer construction at household level whereas 13 municipality had not focused on the conducting such orientation programs . This in turn shows that the household level had some understanding on the need and importance of the earthquake resistant construction practice.

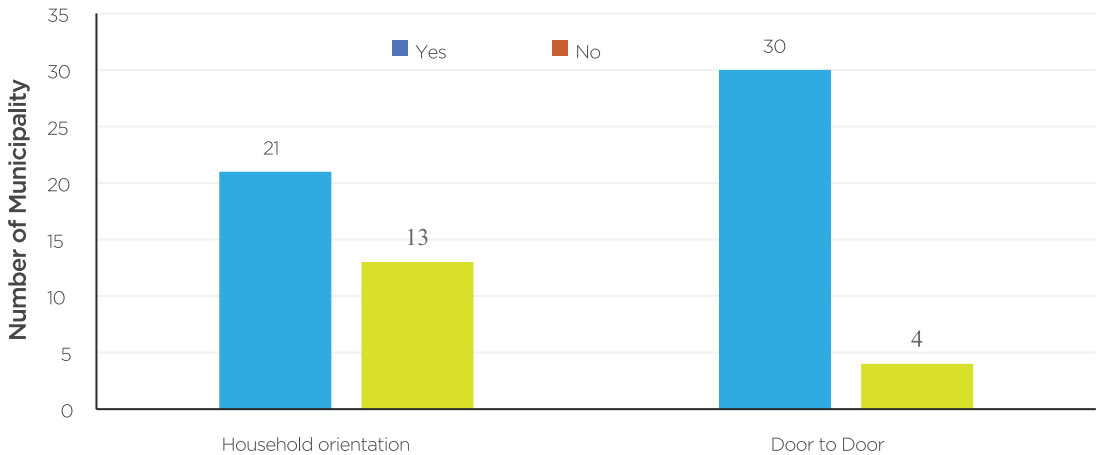
b. Door to Door technical assistance



Technical Assistance coverage map

In contrast to the orientation to the homeowner on earthquake resistant design, it was found that door to door technical assistance was provided by the majority of the municipalities. A large proportion, 30 municipalities (approximately 88%) have provided door to door technical assistance making the municipal population aware on the need and importance of the safer construction practice. The door-to-door technical assistance might have increased in the earthquake affected districts following the reconstruction process. Hence, these kinds of technical activities must be continued as long-term activities for a resilient community.

Technical assistance at household level



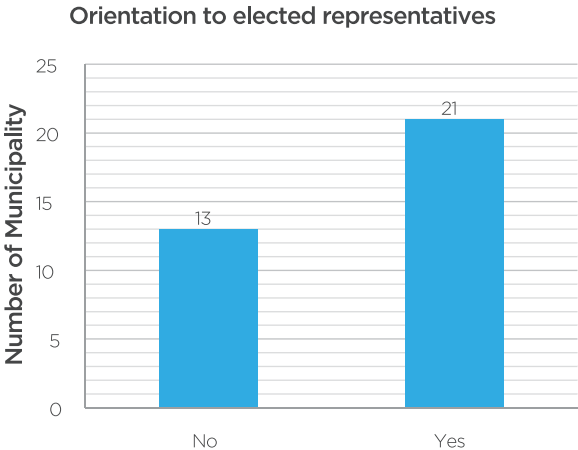
c. Orientation to community leaders on safer construction

Apart from providing orientation and awareness at household level, providing knowledge on safer construction to local leader are of equal importance. The result shows 15 municipalities (approximately 44%) of the surveyed municipalities have provided orientation on safer construction to community leaders while 19 municipalities (56%) have not organized such programs targeting the local leaders. The community leaders have capability to motivate the community people for safer construction practices so, the lack of orientation to such group might lead to continuation of haphazard construction and increase vulnerability during any disaster event.

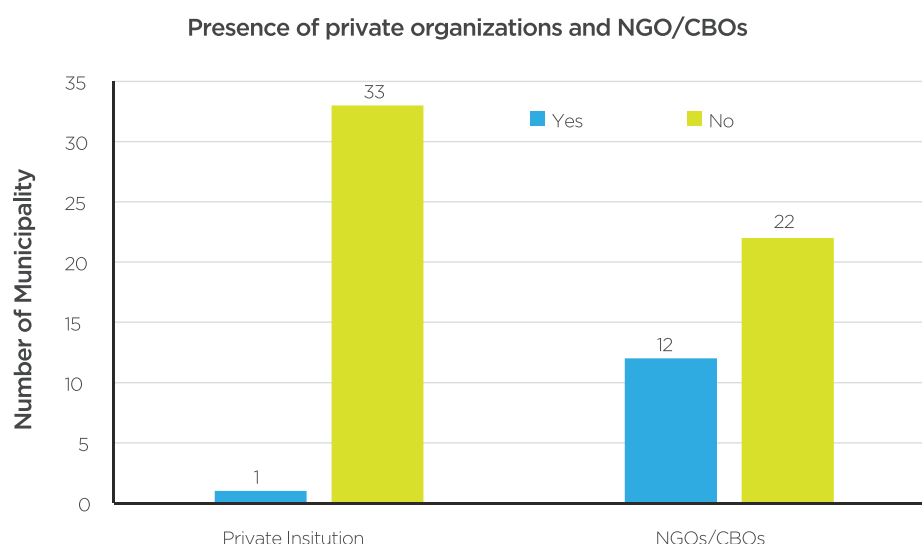


d. Orientation or training on disaster management or urban planning to political leaders and elected representatives

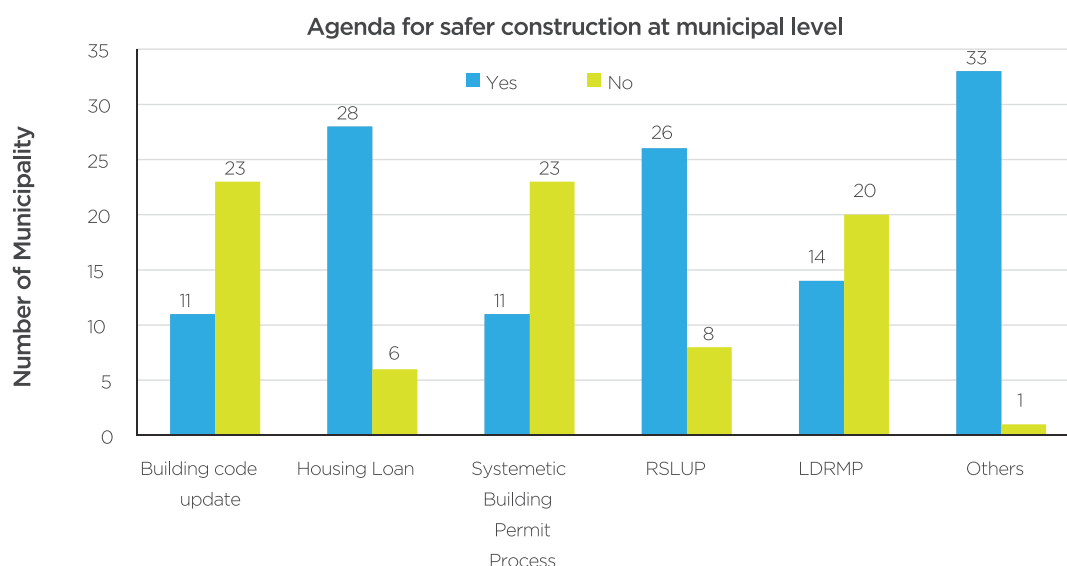
The survey shows 21 municipalities (61.8%) have provided either orientation or training on disaster management or urban planning to political leaders and elected representatives in the past whereas 13 municipalities have not organized any such program as capacity building initiatives for the elected representative. Orientation/training on disaster risk reduction and management to political leaders and local elected representation plays a vital role as they are responsible for the strategic planning and policy making so, these training and awareness on DRR must be provided to such group for sustainable and effective risk reduction.



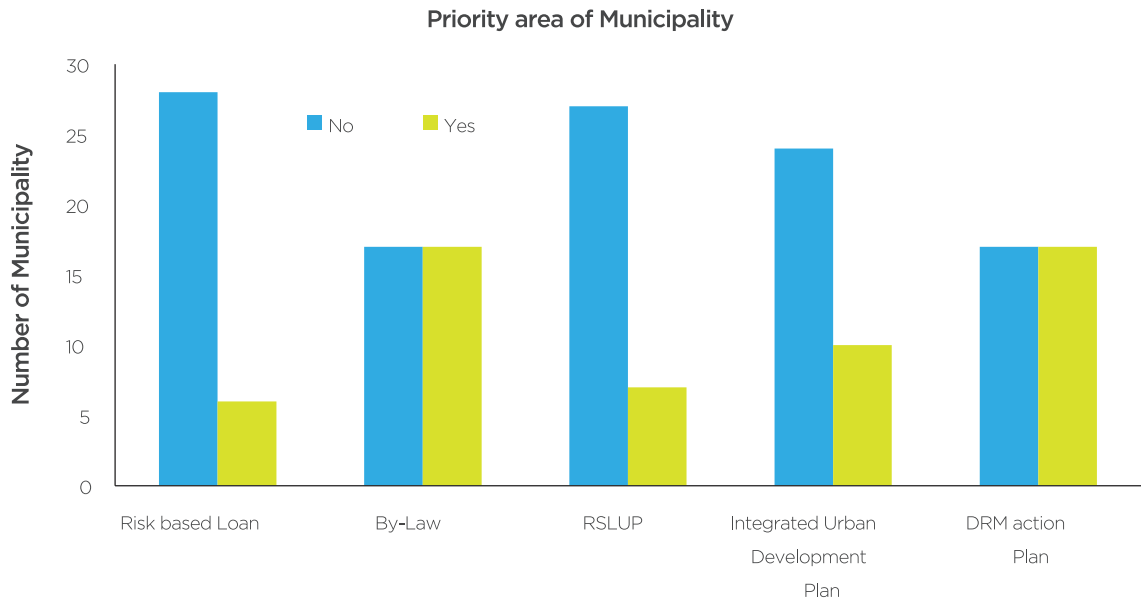
To further understand the reason for having lower number of training/orientations organized on the safer construction or disaster risk reduction, we further investigated the number of private organizations presence in each municipality responsible for organizing such training. From the survey, it has been found that only 1 out of 34 municipalities have presence of private institutions to provide training on earthquake resistant construction to engineers, masons, and petty contractors. Besides private institutions, various NGOs and CBOs have provided training to engineers, masons, and petty contractors on safer construction. It was found that in 12 municipality (35.29%) NGOs and CBOs have provided training to engineers, masons, and petty contractors while in 22 municipalities are remaining to conduct training within their area. The lack of organization/institutes at municipality level specifically providing orientation and awareness program might be one of the major reasons behind low awareness of safer construction and its importance at community level.



Apart from the orientation and training, several interaction meetings or workshops have been organized in municipalities on safer construction. The interaction program focused on several agendas including building permit, building code, DRR and many more. From the survey it was observed that out of 34 only 11 municipalities have conducted interaction program focusing on building code, building permit system which indicate the low preference of these subject within the municipality. The other programs such as housing loan has been conducted in 28 municipality followed by Risk Sensitive Land Use Planning (26 Municipalities), and Local Disaster Risk Management and Planning (14 municipalities). The result further indicates the orientation/training program focusing on building code and building permit system is needed in most of the municipalities.



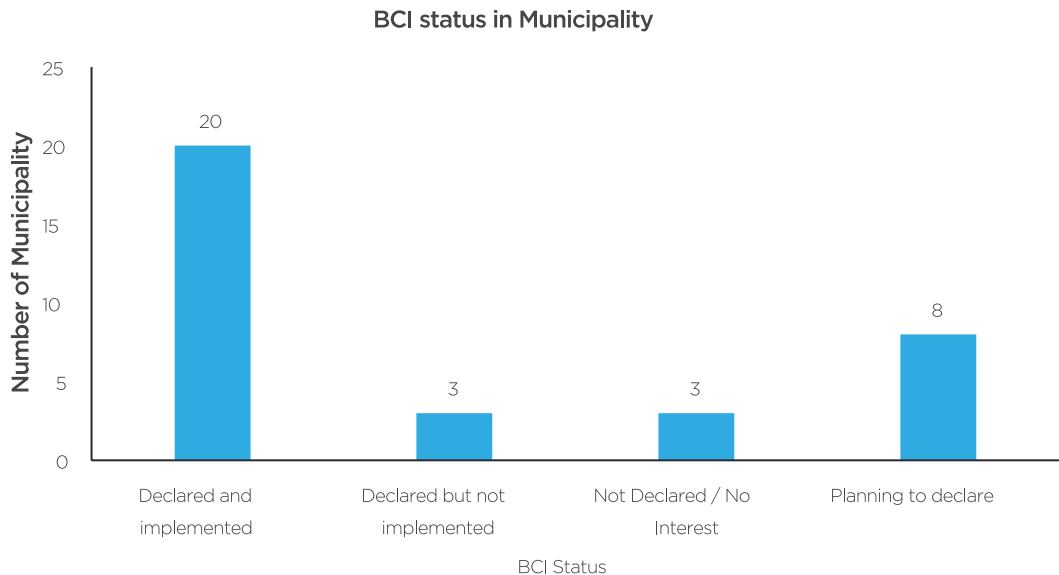
To understand the working modality of the municipality and its priority area the study further investigated the presence of several action plans. In the study it was found that the presence of risk-based housing loan (28 municipalities), Risk Sensitive Land-Use Planning (27 Municipalities), Integrated Urban Development Plan (24 municipalities) was found whereas the presence of Disaster Risk Management plan and Municipal by-laws were found in only 17 municipalities (50%) of the study area. This might indicate that the DRM plan and by-laws are still not among the top priorities of the municipalities even though the risk of hazard is high and increases risk of casualties and huge economic loss.



As per the Urban planning and Development Act 2072, the by-laws are set of rules and regulation framed by municipal planning authorities covering the requirements of building such as FAR: Floor Area Ratio, Right of Way, ensuring safety of the public through open spaces, minimum size of rooms and height and area limitation and many more. The effective implementation of the building by-laws can tackle important issues such as haphazard construction, narrow streets which causes hindrance during any hazards, building devoid of proper lighting and ventilation and so on. Thus, by-laws being one of the most important tools for the physical planning of any municipalities, advocacy for implementation of by-laws has been observed as very crucial in context of Nepal.

3.4.3 Status of Building Code Implementation

Several past researchers have identified that structural failure of buildings has been the single largest cause of casualties and economic losses resulting from earthquake as the number of non-engineered buildings significantly outweighs the number of engineered building in Nepal. For promoting safer and resilient community and nation, construction of safer building acts as one of the most important aspect to determine the risk reduction. Such safer building can only be constructed by following the building code. Thus, understanding the status of the building code implementation is of utmost importance.



Here, from the study it was found that 20 municipalities (approximately 59%) among the surveyed have already declared and implemented the building code, followed by 8 municipalities which are planning to declare. It was also found that among the 16 urban municipalities surveyed, 13 have successfully declared and implemented the building code with remaining planning to declare whereas the municipality not showing interest in building code implementation are mostly the rural municipalities. With the increasing rate of new construction in rural areas of Nepal, the implementation of the building code is much needed which was also observed from the damage caused by the 2015 Gorkha earthquake. Strong advocacy is required to make the building code implementation mandatory in all areas. To investigate more on the status of building code implementation in different areas within the municipality, we sub divided the areas into urban, semi-urban and rural area. It was found that most of the municipalities have implemented building code in all the area whereas some focused on only urban and semi-urban area.

Further, when asked on the current status of the building code implementation in their respective municipality, 21 municipalities agreed to the fact that the level of BCI in their area is moderate and many activities from increasing awareness programs, capacity building programs and training is required.

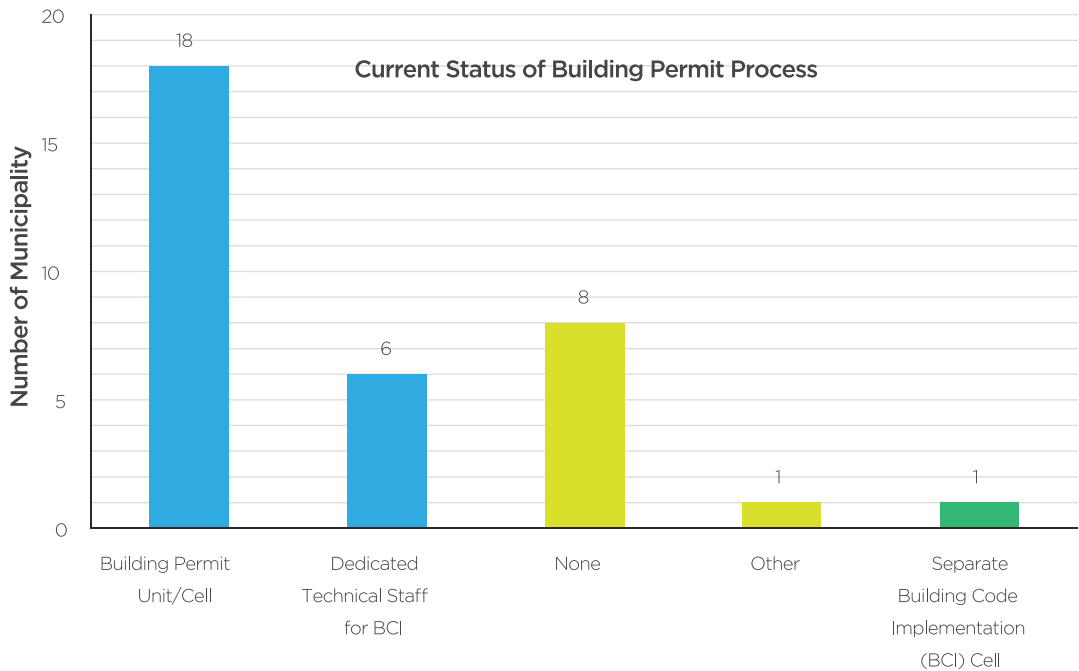
Organizational Involvement

Apart from municipal effort on disaster risk reduction and management including safer construction, several national and international organizations are involved from planning to implementing program and advocating for inclusion of DRR at all level. But to the irony, it was found that only 2 out of 34 municipalities had organization working on BCI. The only organization working for the implementation on the building code was National Society for Earthquake Technology (NSET)- Nepal.

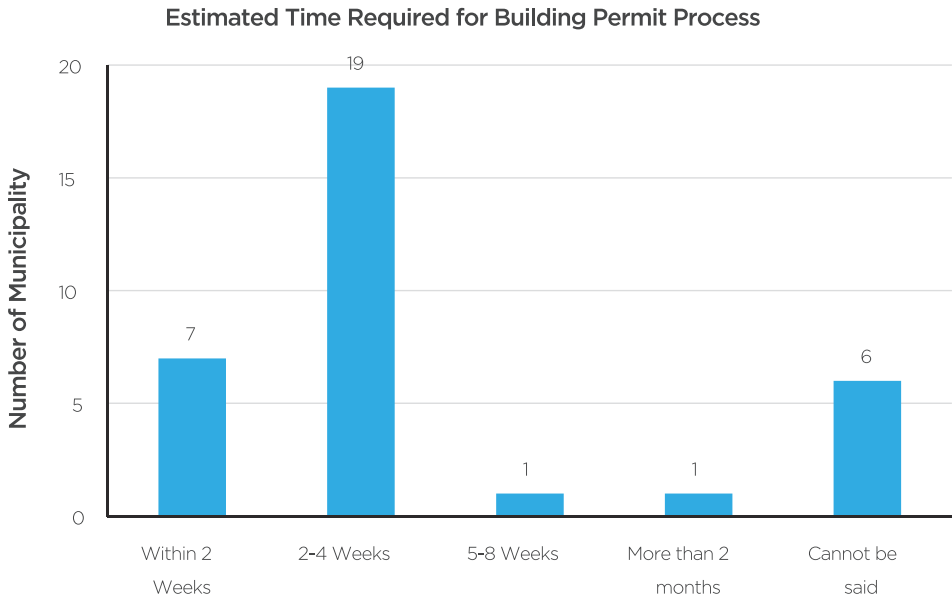
3.4.4 Building Permit Process

To ensure the safer construction following the building code, getting the building permit is the initial and most important process. A building permit is an official approval issued by the municipality/rural municipality that allows to proceed with a new construction, retrofitting, and addition of building storey. The building permit process includes submission of application to the municipality along with the official land registration certificate, tax payment receipt, architectural drawings, structural drawings, electrical and sanitary drawings along with the structural Analysis report as per the National Building Code of Nepal or the Indian Standards Code. Even though getting building permit has been mandatory in most of the municipalities of Nepal, many houses are being constructed without going through the building permit process. There might be several reasons such as lack of landownership document, lack of knowledge on the process, complicated and lengthy process etc. due to which people are not going into the building permit process. To understand the institutionalization status of the building permit process at municipality level, we further asked the about the current status of the building permit process, estimated time for getting the approval, presence of technical staffs etc.

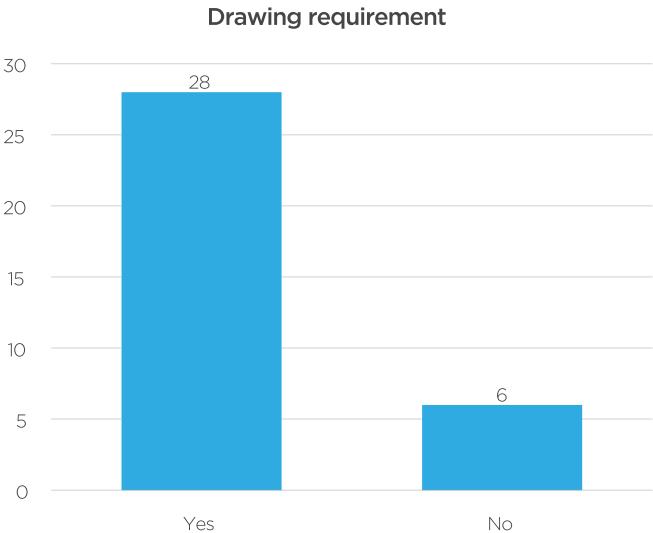
It was found that out of 34 surveyed municipality, 18 municipalities had separate building permit cell or unit to provide service to the local resident. Similarly, 6 municipalities had technical staffs dedicated to providing information or facilitate the building permit system whereas 8 municipalities had no special provision for the building permit process.



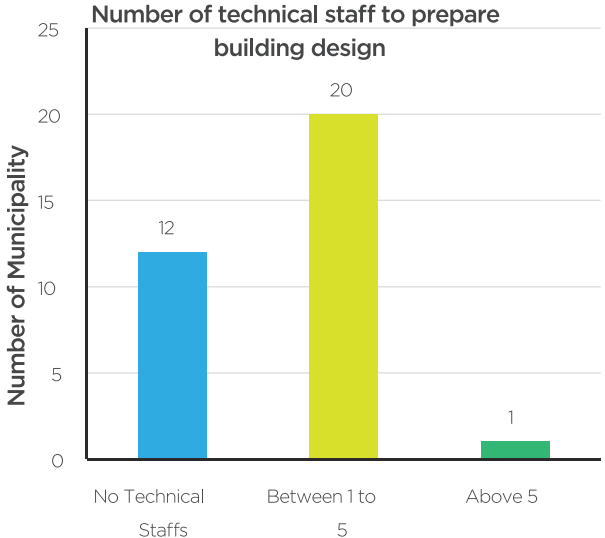
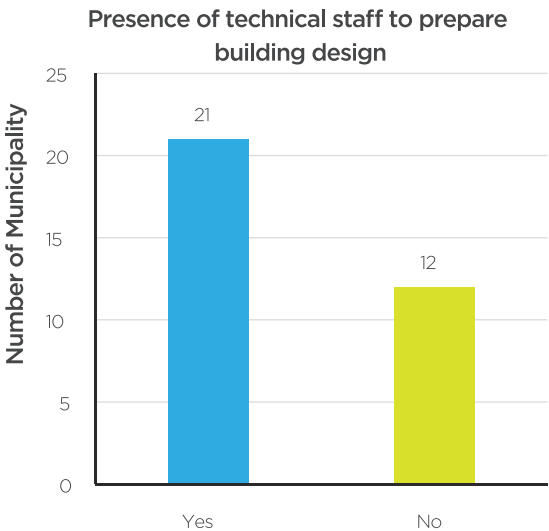
As per the Technical Inspection Guideline for Housing Reconstruction, it usually takes 17 days for simplified process for building permit if all the administrative and technical requirements are fulfilled. From our study it was found that most of the municipality i.e., 26 (nearly 87%) issue the building permit within the time period as specified by technical inspection guideline whereas 6 municipality were not able to specify the time taken for an individual to get the building permit which would eventually lead to the delay in the construction.



Depending on the municipality, the required documents for the building can vary. For example: Kathmandu Metropolitan City has mandatory rule of including electrical and sanitary drawings and a structural analysis report along with other documents. Regarding the need of structural drawing for building class "C" (Building with plinth area of up to 1000 sq. ft , with three floors including the ground floor or with structural span of up to 4.5 meters) & and detail designs for class "A" (Modern building) and "B" (Buildings with plinth area of more than 1000 sq. ft, with more than three floors including the ground floor or with structural span of more than 4.5 meters) buildings for the building permit process, it was found majority of the municipalities (28 municipalities) require such drawings to ensure the safer construction.



To further get the scenario of the competency of the available technical staff to design category B type building, it was found that only 21 municipalities had technical capacity of preparing design of such structure whereas 13 municipality did not have presence of such technical staffs. Among the municipality having technical staff available to prepare design of category B building type, we further explored the number of such available staffs, where it was found that majority of the municipalities (20 municipalities) had number of staffs ranging between 1 and 5 and only 1 municipality had more than 5 technical staffs. The data further suggests the need of more engineers and consultants as the construction of Category B type buildings are increasing especially in the urban and financial centers of Nepal.

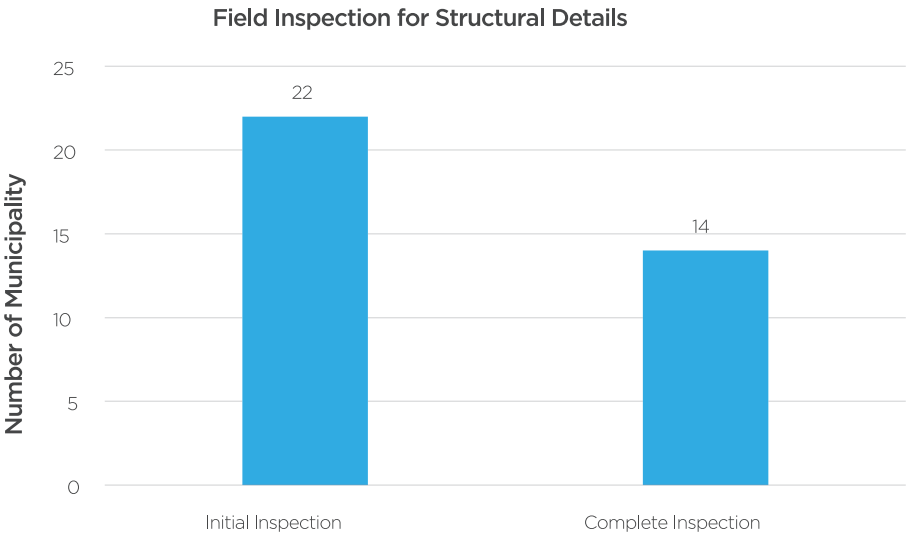
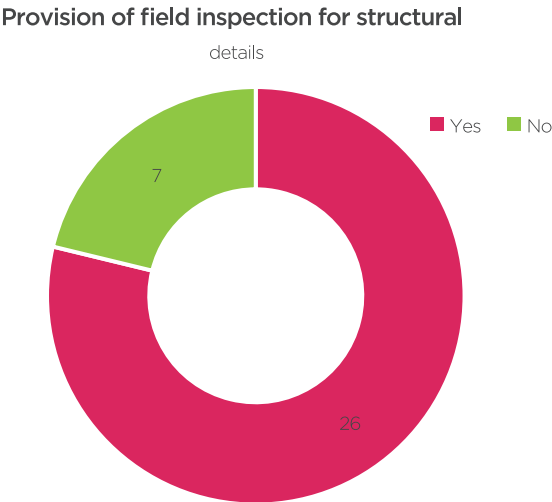


Field Inspection

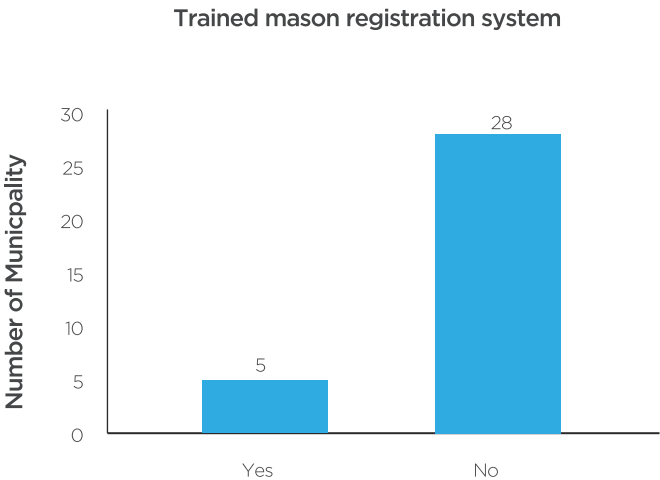
Field inspection is most important and crucial step in any construction as it is conducted to validate the construction as per the building permit and provide technical assistance to the houseowner. Of the 34 municipalities, 26 municipalities (79%) have the provision of the

field inspection for structural details. Those municipalities having the provision for the field inspection were further asked on the level of field inspection conducted by them.

Among the 26 municipalities having provision of field inspection, 22 municipalities have been found to conduct the initial level inspection i.e., at foundation level but the number decreases and only 14 among them continue the inspection to the final level including inspection at all levels i.e., at foundation, plinth level, at each storey and final completion . The remaining municipalities were found to conduct field inspection at any level such as after completion of storey or plinth level. The data further indicates that 7 municipalities are not monitoring and validating the construction process as per the building permit system. The lack of field inspection from municipality technical staffs might result in modification in the construction increasing the vulnerability of the structure



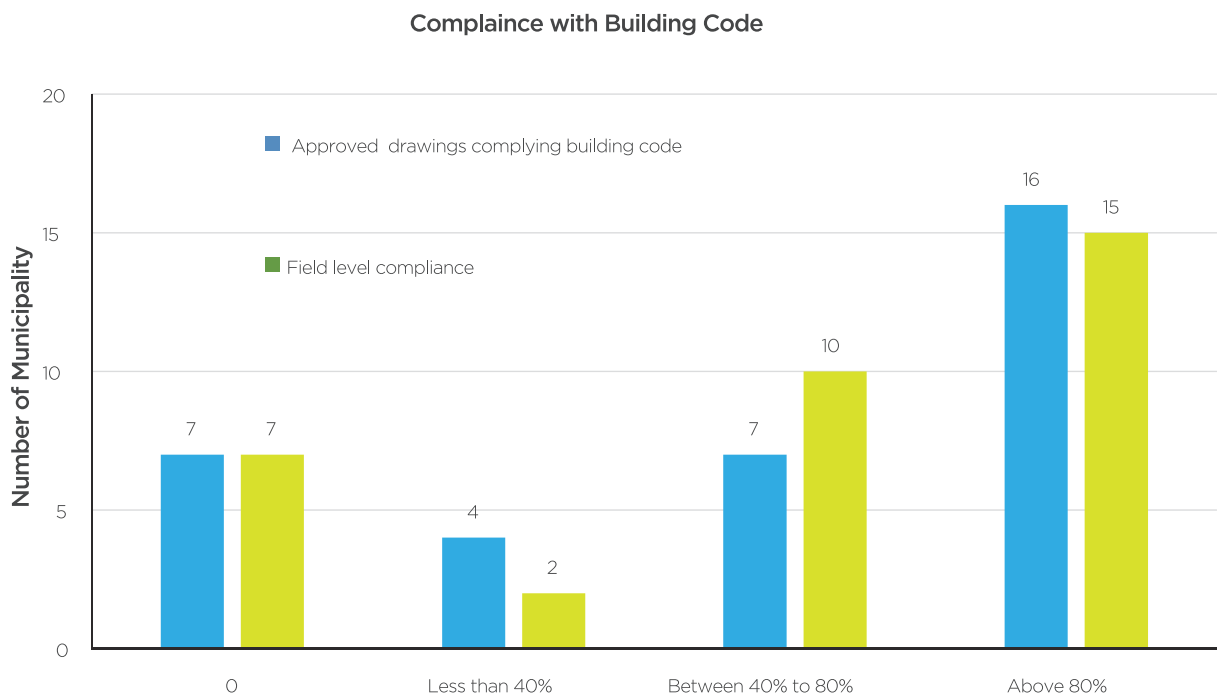
Most of the residential buildings are constructed under the guidance and involvement of a head-mason and his team. The construction practice in Nepal are usually owner-driven and where the houseowner have tendency to construct house as per the suggestion and direction of head masons or petty contractor. It is believed that during the deficiency or unavailability of the municipal technical staff, the trained mason is technically sound in guiding other masons at each step of building construction. Following this scenario, we asked if there are any provision of municipality preparing a roster of trained mason. Unfortunately, there are only 15% of the municipalities (i.e., 5 out of 34) registering the list of the trained mason. Regular update and maintenance



of the database of trained mason in the municipal database system is important as they are the resources for the municipality that can be fully mobilized in regular construction process and post disaster scenario, especially in the rehabilitation and the reconstruction phase.

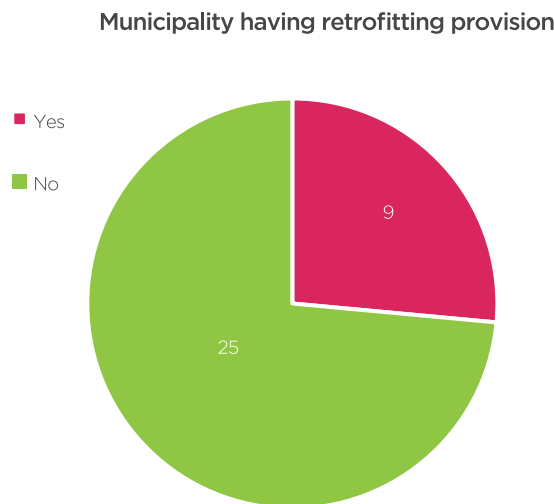
Building Drawing Compliance with the Building Code

Among the number of approved % of drawings complying with building code between 80% to 100% is 16 whereas still 10 of the surveyed municipalities have compliance rate less than 80%. Regarding the field level compliance only 15 municipalities had compliance of more than 80%. The increasing number of technical assistance activities such as door to door technical assistance, community orientation might have influence in increasing in the field compliance rate. Further comparing the data in terms of rural and urban municipalities, lower compliance rate was found be in rural municipalities. This result might indicate the trend of unsafe building construction in Nepal and high risk during future earthquake. The critical need of technical staff for the preparation of the building design as well as field inspection has been identified by this study.



Municipal Provision for Retrofitting

Seismic retrofitting is still considered new and challenging process in the context of Nepal. Retrofitting is required for houses that are vulnerable to seismic waves or slightly damaged. Retrofitting of the existing structures may reduce the overall damage caused by the earthquake. However, many of the surveyed municipalities do not have the retrofitting provision or they are not aware of the importance of retrofitting. It was found that majority of the municipalities i.e., 25 out of 34 had no retrofitting provision and only 9 municipalities, i.e., 26% of the municipalities have considered retrofitting option for the structural safety. Compared to rural, urban



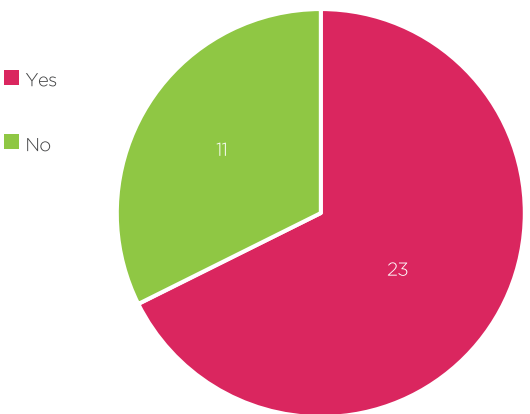
municipalities have recognized the importance of retrofitting for improving the resistance levels of the residential structures to the earthquake. This is clearly demonstrated through the survey data; as out of the 9 municipalities having retrofitting provision, 6 are urban and 3 are rural municipalities.

As per the PDNA, many buildings had suffered partial damage during the 2015 Gorkha Earthquake and were categorized under retrofitting but due to technical deficiency, awareness and government policies, the progress in retrofitting is almost negligible as compared to reconstruction. This indicates the important of the seismic retrofitting guideline and policies not only at local level but also at provincial and national level.

Technical Requirement

The role of technical person is very critical for a safer construction which usually involves preparing economically feasible designing and execution assuring combination of accuracy, safety, durability, speed, simplicity, efficiency. Technical person requires academic training, examination of competence, and professional ethics. Out of 34 the surveyed municipalities, only 23 municipalities have specified minimum qualification of the technical staff for preparing the building design. Among these municipalities, 17 municipalities responded that they have set minimum qualification for only engineers whereas 6 had qualification specified for both engineers and sub engineers.

Minimum qualification of technical staffs for building drawings



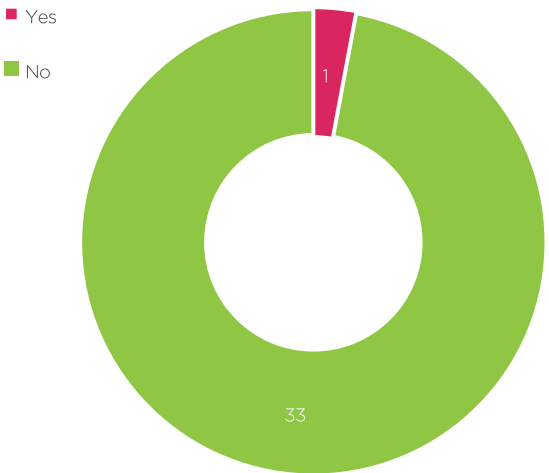
As the rate of construction are increasing in both urban and rural municipalities, only the municipal engineers are not able to prepare the design for the planned construction within their area, so the involvement of consultants, architects and planners also needs to be considered. It was found that 50% of the municipalities (17 out of 34) had organized system of registration of construction practitioners, e.g., architects, engineers, town planners and construction supervisors in municipalities.

3.4.5 Municipal Action Plan

Computer software or mobile application

The e-BPS system is a very innovative system where all the documents and information are uploaded electronically and then forwarded for the building permit process. The e-BPS system has been implemented in all municipalities within Kathmandu Metropolitan City in the fiscal year 2073-74. The use of computer software or mobile application for the processing the building permits and maintaining the current building records in the surveyed municipalities is nominal. The BCI study shows that only 1 (3%) urban municipality has practiced computer software or mobile application for this purpose. This also pointed out that municipalities should

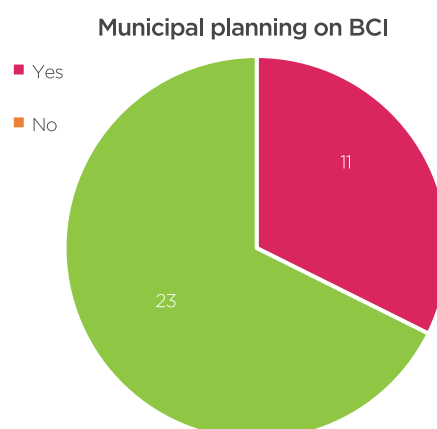
Computer software or mobile application



replace the paper-based method with digital technology, so that citizens can have access to building permits easily and promptly.

Building Code

In Nepal, municipality is the responsible authority for implementing National Building Code at local level. The municipality not only issues the building permit, but also monitor and evaluate the current status of the building construction. However, majority of the surveyed municipality doesn't have any plan for building code implementation. Only 11 (32%) out of 34 municipalities have plans to implement the building code into their building permit process. The lack of planning for building code implementation shows the need of capacitating the municipal authorities on the need and importance of building code implementation.



Effective implementation of building code is one of the most practical ways to eliminate risk of damages caused due to seismic hazard. Among the municipality planning on implementing building code, detail question of the kind of program planned by the municipalities were asked. Majority of the municipalities are planning for technical training to masons and engineer followed by the awareness program. To ensure the implementation of building code and its awareness municipalities have also been planning several activities and workshop. Majority of the municipalities have conducted discussion and orientation program followed by experience sharing and learning through visit to different municipalities.

Budget for Building Code Implementation

The table 4 shows the average annual budget of the surveyed municipalities and the allocated budget for the building code implementation. The minimum budget allocated in for the BCI in both rural and urban municipalities were found to be NRP. 100 (in thousand) whereas the maximum budget allocated in urban was found way higher than rural area. On an average 0.25% and 0.48% of the annual municipal budget of rural and urban are allocated for the BCI respectively. The allocated budget is utilized for the BCI related activities including training, orientation/workshop, publications, and relevant programs.

Municipality	Minimum		Maximum	
	Annual Municipal Budget (In thousand)	Budget allocated for BCI (In thousand)	Annual Municipal Budget (In thousand)	Budget allocated for BCI (In thousand)
Rural	30,000	100	623,100	1,000
Urban	78,000	100	1,700,000	14,211
Analysis for budget allocation on BCI	Minimum	Maximum	Average	Standard Deviation
Rural	0.33%	0.16%	0.25%	0.12%
Urban	0.13%	0.84%	0.48%	0.50%
Annual Income from Building Permit	Minimum (In thousand)		Maximum (In thousand)	
Rural	100		12,365	
Urban	100		28,000	

Here, the standard deviation for budget allocation of BCI in rural was found to be smaller than urban municipalities. Due to the limitation of the municipal sample size and difference in the sample size of rural and urban municipality, the exact relation between the BCI budget allocated couldn't be derived appropriately. Hence, more detailed survey is recommended for this matter.

Table 4: Annual municipal budget for the BCI implementation and annual income from building permit system

Similarly, looking at the revenue collected by the municipality from the building permit process, though the minimum annual income in both rural and urban municipalities are NRP. 100 (in thousand) but the maximum revenue of urban municipalities is significantly higher than that of compared to the rural municipality.

3.5 Analysis and interpretation

3.5.1 Number of municipal technical staff Vs Houses construction per year

Regarding the availability of municipal technical staff, on an average 9 technical staff were found in rural areas and 12 in urban area as indicated in Table 5. Further looking at the houses constructed per year, on an average 1 technical staff was found to be responsible for the supervision of 70 houses per year in rural area whereas in urban area 1 technical staff is responsible for 47 houses per year. This in turn indicates the necessity of the technical staffs in municipalities especially in the rural areas where the number of constructions is increasing every year.

Municipal	Number of Technical Staff	House Constructed Per Year	Number of houses supervised in per year
Rural	9	626	70
Urban	12	564	47

Table 5: Number of technical staffs versus houses constructed per year

3.5.2 Mason and petty contractor Vs Houses construction per year

Further, Table 6 shows the number of mason/petty contractor present in the rural and urban municipalities versus the average number of houses constructed per year. It was observed that, even though the number of houses constructed per year is higher in rural area, the number of masons is less.

Table 6: Number of mason/petty contractor

Municipality	Average number of mason/petty contractor	Average house constructed per year
Rural	344	626
Urban	472	564

It was found that on an average there might be presence of 3 technical consultant in rural municipalities and 29 in urban municipalities (Table 7). Further, comparing the result with the number of houses constructed within the municipalities, it was found that on an average approximately 1 consultant in rural municipalities is responsible from preparing drawing to field monitoring of 205 houses whereas in urban area the number is significantly low where 1 consultant is responsible for only 20 houses.

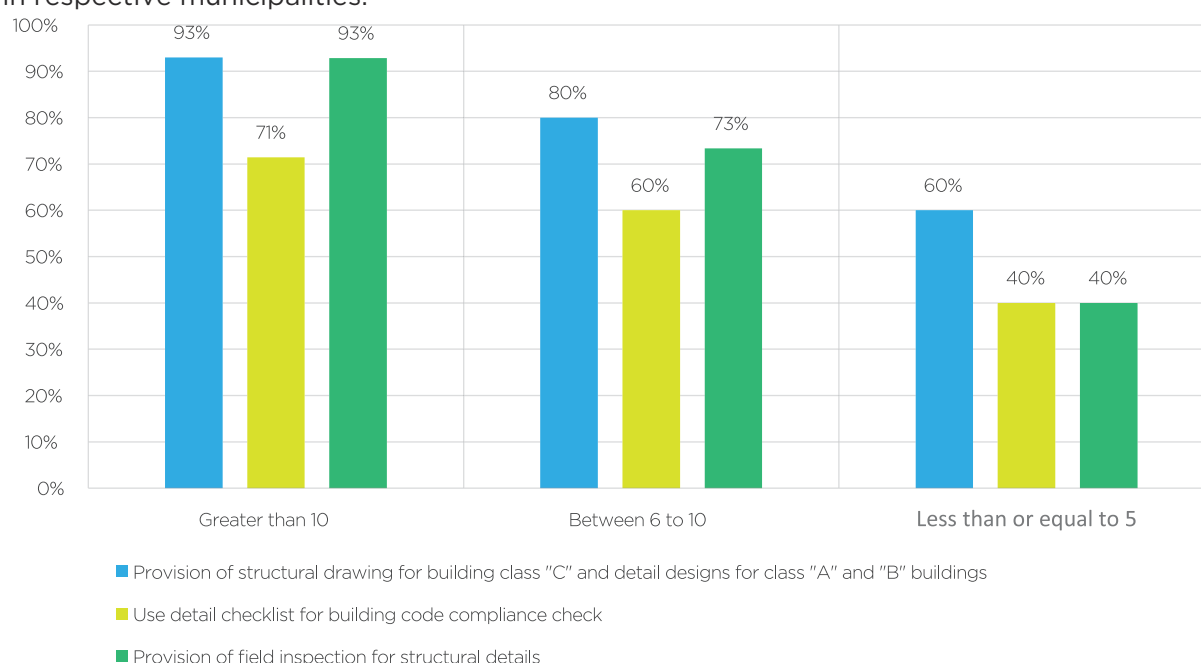
Types	Average Consultant	Average House Constructed per year	Number of houses supervised in per year
Rural Municipality	3	626	205
Urban Municipality	29	564	20

Following the 2015 Gorkha earthquake and government post-earthquake reconstruction program, the construction rate in both urban and rural area of the 32 earthquake affected districts has increased within few years. As the survey was conducted during the reconstruction time frame, the number of houses construction might differ from the regular construction period.

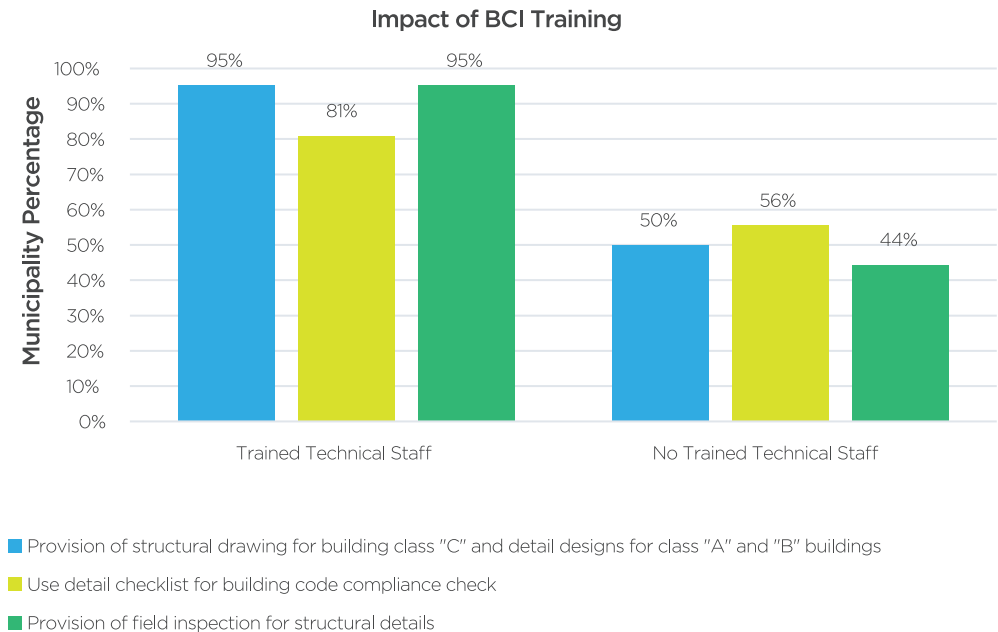
Table 7: Number of consultant engineers and consultant firm

3.5.3 Correlation between the number of municipal technical staffs and BCI

Considering the number of municipal technical staff and the provision of BCI in the study area, it was found that higher the number of technical staffs higher is the inclusion of BCI components. In the municipalities with presence of more than 10 technical staffs, 93% of such municipalities have provision of structural drawing for building class “C” and detail designs for class “A” and “B” building and provision of field inspection for structural details and 71% have provision of using the detail compliance checklist. This further indicates that the status of building code implementation might be influenced by the number of technical staff present in respective municipalities.



To understand the impact of the training on BCI in the municipalities, it was found that municipalities with trained technical staffs have higher inclusion of BCI components including the provision of structural drawing for building class “C” and detail designs for class “A” and “B” building and provision of field inspection for structural details and provision of using the detail compliance checklist. This further indicates the need and advantage of BCI training to the municipal technical staffs.



3.6 Identification of the most critical issues and Challenges

In terms of influence of engineers and professional also shows significant difference in urban and rural area. In rural area the presence of technical person is almost negligible, and the head mason lead the overall construction process whereas in urban area though there is involvement of technical person from designing to final construction phase, but their presence during ongoing construction stage is still very less.

Effective enforcement of earthquake resistant building codes and control system can reduce the human casualties and loss significantly. Realizing this fact, the National Building Code (NBC) has been developed in 1992 to improve the structural safety of buildings and reduce the impact of earthquakes in life and livelihood of the people. It is almost 12 years since the code is developed but implementation has not been as effective. Implementation of NBC is not only a technical issue but also includes social, legal, and institutional aspect such as lack of awareness, lack of institutional mechanism for implementation and inadequate capacity of implementing authorities. Despite of many constraints and challenges, implementation of NBC is a key factor in reducing the disaster risk and hence is very important. Therefore, there is a need to go ahead with implementation of NBC with various pragmatic and innovative ideas.

4. Conclusion

Based on the results and finding of the study several gaps were found in the inclusion of National Building Code and Building permit system. The study indicated the necessity of the technical staffs in municipalities especially in the rural areas where the number of constructions is increasing every year. Despite the government making the implementation of building code mandatory in all the municipalities of Nepal, there are still huge gap in the level of implementation are majority of the municipalities have yet to incorporate building code in their respective area. The gap was also identified in the number of training and awareness program regarding the building permit system, disaster risk reduction and management, as majority of those were targeted for technical staffs and masons only. Irrespective of training and capacity building program being targeted to the technical staffs and mason, there were still huge gaps in the number of trained and untrained human resource. It was also found that majority of the municipality had not maintained any database of the technical staffs, mason, consultants available in their area. Regarding the building permit system as well as the municipal programs including trainings were focused on the new construction rather than strengthening the capacity of the existing building. So, the need for inclusion of retrofitting process in the building permit system as well as local government program must be done to ensure the reduction of damage of old structure during future earthquake. The role of local government in the implementation of the building code is very critical and important for risk reduction but unfortunately program focusing on building safety, consistency in the compliance monitoring is not among the top priorities of the municipalities even though the Nepal lies in the high risk of hazard and causes casualties and huge economic loss. With the ongoing post-earthquake reconstruction, awareness program such as household orientations, door to door technical assistance on safer construction has significantly increased along with the publication of technical guidelines, policies which needs to be incorporated within the regular activities of the local government depending on the need and relevancy to develop a safer and resilient community.

4.1 Recommendation and Way forward

Based on the finding of the studies, few recommendations and way forward are provided below.

- The placement of technical staffs especially in the rural municipalities of Nepal must be ensured to encourage safer construction practices. In case of deficiency in number of technical staffs, process for the deployment of staffs must be done to fulfill the demand as well as the vacant seats as per requirement.
- Regular capacity building program to different target groups including engineers, masons, social mobilizer, community leader and local elected representative on building code must be included in the municipality action plan.
- Awareness programs and orientation on importance of the BCI and safer construction must be conducted from household level to local government including municipal staffs.
- Advocate for the mandatory implementation of building code and by-laws in all rural and urban municipalities to ensure safer and resilient construction practice.
- A procedure of surveying and recording the number of technical staffs, trained masons, consulting firms is recommended. A separate database or roster of such human resource must be prepared at municipal level to plan their mobilization as per need.

- The local government should have a strong mechanism and consistency in the monitoring and compliance check of the new construction.
- Inclusion of field inspection and compliance checking in the building permit system is recommended for those municipalities which don't have field inspection provision.
- Provision of retrofitting must be incorporated in the building permit process of the municipality.
- Inclusion of retrofitting process on the local government program must be done to ensure the reduction of damage of old structure during future earthquake.
- The technical guidelines, policies, and program as part of 2015 post-earthquake recovery and reconstruction should be incorporated as part the regular municipal program depending on the requirement and for long-term sustainability and resilient community.

5. Annexes

Annex 1 - Common building typology in Nepal (DUDBC, 2009)²

Building Types	Description
Low strength masonry (adobe or brick or stone in mud mortar).	Adobe buildings are constructed using sun-dried bricks (earthen) with mud mortar for the construction of structural walls. The wall thickness is usually more than 350 mm. Stone-masonry buildings are constructed using dressed or undressed stones with mud mortar. These types of buildings have generally flexible floors and roof. Low strength brick masonry includes brick masonry buildings with fired bricks in mud mortar.
Masonry (brick or stone in cement mortar)	Masonry buildings are the brick masonry buildings with fired bricks in cement or lime mortar or stone-masonry buildings using dressed or undressed stones with cement mortar.
Non-engineered Reinforced Concrete Moment Resisting-Frame Buildings	Non-engineered reinforced concrete buildings typically have reinforced concrete frames and unreinforced brick masonry infill in cement mortar. The thickness of infill walls is 230mm (9") or 115mm (4 1/2") and column size is predominantly 9"x9". The prevalent practice in most urban areas of Nepal for the construction of residential and commercial complexes generally falls under this category. These buildings are not structurally designed or supervised by engineers during construction. This category also includes the buildings that have architectural drawings prepared by engineers, but do not have structural designs.
Engineered Reinforced Concrete Moment Resisting-Frame Buildings	These buildings consist of a frame assembly of cast-in-situ concrete beams and columns. Floor and roof framings consist of cast-in-situ concrete slabs. Lateral forces are resisted by concrete moment frames that develop their stiffness through monolithic beam-column connections. These are engineered buildings with structural design and construction supervision by engineers. Some of the newly constructed reinforced concrete buildings are of this type.
Others	Wooden buildings, and hybrid buildings that combine multiple load-bearing masonry are also found in the Kathmandu Valley and other part of the country.

² Seismic Vulnerability Evaluation Guideline for Private and Public Buildings, Part II: Post Disaster Damage Assessment, 2009

Annex 2 - Existing National Building Code in Nepal, DUDBC

Code Number	Code Title
NBC 000:1994	Requirements for State-of-the-Art-Design. An Introduction https://www.hrrpnepal.org/uploads/media/SjNqnQ1tbaLEciF7m0fM_2017_02_23.pdf
NBC 101:1994	Material Specifications https://www.hrrpnepal.org/uploads/media/qvkPODA35sVcYCMrdOnN_2017_02_23.pdf
NBC 102:1994	Unit Weight of Materials https://www.hrrpnepal.org/uploads/media/7IfUQNbKV5gqHOSmJxpA_2017_02_23.pdf
NBC 103:1994	Occupancy Load(Imposed load) https://www.dudbc.gov.np/uploads/default/files/ba86946f1a8af674d1c6f5a151fa2ead.pdf
NBC 104:1994	Wind Load https://www.dudbc.gov.np/uploads/default/files/88684e75d348443c04e3778316257509.pdf
NBC 105:2020	Seismic Design of Buildings in Nepal https://www.dudbc.gov.np/uploads/default/files/e91f6292eb11fb9872760af469311bce.pdf
NBC 106:1994	Snow Load https://www.dudbc.gov.np/uploads/default/files/87de7d35b6ff580da67103b85da42122.pdf
NBC 107:1994	Provisional Recommendation on Fire Safety https://www.dudbc.gov.np/uploads/default/files/998b31e2549c235445c152204355bc90.pdf
NBC 108:1994	Site Consideration for Seismic Hazard https://www.dudbc.gov.np/uploads/default/files/d5b1c95f36d26a002d468c3e32059d7f.pdf
NBC 109:1994	Masonry (Unreinforced) https://www.dudbc.gov.np/uploads/default/files/5f8e64743168c770bbbf7e4d02d38918.pdf
NBC 110:1994	Plain and Reinforced Concrete https://www.dudbc.gov.np/uploads/default/files/b8e8f3b62d18e2fd94ee0942fdd34.pdf
NBC 111:1994	Steel https://www.dudbc.gov.np/uploads/default/files/b8e8f3b62d18e2fd94ee0942fdd34.pdf
NBC 112:1994	Timber https://www.dudbc.gov.np/uploads/default/files/09c6f80a5cf823991cd05cb90a1ab677.pdf
NBC 113:1994	Aluminum https://www.dudbc.gov.np/uploads/default/files/fc99d9107e10e08b9d91e7b568207b7e.pdf

NBC 114:1994	Construction Safety https://www.dudbc.gov.np/uploads/default/files/c4f2d67c03bc00d4e9a280f2223392e4.pdf
NBC 201:1994	Mandatory Rules of Thumb Reinforced Concrete Buildings with Masonry Infill https://www.dudbc.gov.np/uploads/default/files/30f57e54649ba9b6e03dbdef12929ea5.pdf
NBC 202:2015	Mandatory Rules of Thumb Load Bearing Masonry https://www.dudbc.gov.np/uploads/default/files/9576109c33df2d9d5cabd8a8ca6a2ce5.pdf
NBC 203:2015	Guidelines for Earthquake Resistant Building Construction (Low Strength Masonry) https://www.dudbc.gov.np/uploads/default/files/b8e8f70594055e66f43b739ed48ae057.pdf
NBC 204:2015	Guidelines for Earthquake Resistant Building Construction (Earthen Building (EB)) https://www.dudbc.gov.np/uploads/default/files/52e69c2ecf9c35bb64a8f22884dc4f8c.pdf
NBC 205:1994	Mandatory Rules of Thumb Reinforced Concrete Buildings without Masonry Infill https://www.dudbc.gov.np/uploads/default/files/b3075826fc5515675b9f0e7d78d617cf.pdf
NBC 206:2015	Architectural Design Requirements https://www.dudbc.gov.np/uploads/default/files/eefb050a5a45f1e1ce05a75875ad63bc.pdf
NBC 207:2003	Electrical Design Requirements for Public Buildings https://www.dudbc.gov.np/uploads/default/files/555f2944e317eebb87a19d7da2af4910.pdf
NBC 208:2003	Sanitary and Plumbing Design Requirements https://www.dudbc.gov.np/uploads/default/files/5516d1570820fec0d862760faeac47a9.pdf

Annex 3 - Survey Questionnaire

7/1/2020

Municipality Baseline Information Collection Checklist

Municipality Baseline Information Collection Checklist

यो सर्वेक्षण आवास पुनर्स्थापना तथा पुनर्निर्माण समन्वय मञ्च(HOUSING RECOVERY AND RECONSTRUCTION PLATFORM)बाट गरिएको हो । यसको मुख्य उद्देश्य नगरपालिका/गाउँपालिकामा भवन निर्माण निर्माण संहिता कार्यान्वयन को अवस्था, नगरपालिका/ गाउँपालिकामा भवन निर्माण संहिता सम्बन्धि चेतनाको स्तर र नगरपालिका/ गाउँपालिकामा भवन निर्माण संहिता कार्यान्वयनको लागि उपलब्ध जनशक्तिको अवस्था बारे थाहा पाउनु हो । यो सर्वेक्षणको उद्देश्य वाहेक अरु कुनै प्रयोजनको लागि सर्वेक्षणबाट आएका तथ्याङ्क प्रयोग गरिने छैन ।

Read out to the respondent

के तपाईं यो सर्वेमा भाग लिन ईच्छुक हुनु हुन्छ ? *

Respondent's CONSENT

☒ छ

☐ छैन

क. उत्तरदाताको बिबरण

उत्तरदाताको नाम: *

उत्तरदाताको पद: *

उत्तरदाताको संपर्क नं.: *

प्रदेशको नाम *

- ☐ Province 1
- ☐ Province 2
- ☐ Bagmati
- ☐ Gandaki
- ☐ Province 5
- ☐ Karnali
- ☐ Sudurpaschim

जिल्लाको नाम *

गाउँ/नगरपालिकाको नाम *

गाउँ/नगरपालिका घोषणा भएको बर्ष: *

Please enter the date in BS.

गाउँ/नगरपालिकाले ओगटेको क्षेत्रफल: *

Please enter the area in sq. km.

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1/15

7/1/2020

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गाँऊ/नगरपालिकाको कुल जनसंख्या:	*
स्रोत (जनसंख्या):	*

ख . घर निर्माणको अवस्था

हाल रहेका जम्मा घरहरू: <i>Please enter here</i>	*	गत वर्ष बनेका नयाँ घरहरू: <i>Please enter here</i>	*	नक्सा पास नगरी बनेका नयाँ घरहरू औसत प्रतिशत (%) प्रति वर्ष <i>नक्सा पास गरेको र नगरेको गरेर १०० % हुनु पर्ने</i>	*	नक्सा पास गरेर बनेका नयाँ घरहरू औसत प्रतिशत (%) प्रति वर्ष <i>नक्सा पास गरेको र नगरेको गरेर १०० % हुनु पर्ने</i>	*
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ग . घरका प्रकारहरू

जम्मा घरहरूमध्ये आर सी सी (RCC) घरहरू <i>Please enter in number</i>					
जम्मा घरहरूमध्ये गारोवाला (MASONRY) घरहरू <i>Please enter in number</i>					
हरेक वर्ष निर्माण हुने आर. सी. सी. (RCC) घरहरू: औसत प्रतिशत (%) प्रति वर्ष <i>RCC, SMM and Other should be 100 % in Total</i>	*	हरेक वर्ष निर्माण हुने गारोवाला घरहरू: औसत प्रतिशत (%) प्रति वर्ष <i>RCC, SMM and Other should be 100 % in Total</i>	*	हरेक वर्ष निर्माण हुने अन्य प्रकारका नयाँ घरहरू: औसत प्रतिशत (%) प्रति वर्ष <i>RCC, SMM and Other should be 100 % in Total</i>	*

घ. तला अनुसारको बर्गीकरण (प्रतिशतमा %)

तला अनुसारको बर्गीकरण	३ तला सम्म रहेको घरहरू (प्रतिशतमा %)	३ तला भन्दा माथि रहेको घरहरू (प्रतिशतमा %)
जम्मा आर. सी. सी. (RCC) भवनहरू मध्ये	* <i>Total of 3 storey and more than 3 should be 100 %</i>	* <i>Total of 3 storey and more than 3 should be 100 %</i>

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2/15

7/1/2020

Municipality Baseline Information Collection Checklist

तला अनुसारको बर्गीकरण	२ तला सम्म रहेको घरहरू (प्रतिशतमा %)	२ तला भन्दा माथि रहेको घरहरू (प्रतिशतमा %)
जम्मा गारोवाला भवनहरू मध्ये	* Total of 2 storey and more than 2 should be 100 %	* Total of 2 storey and more than 2 should be 100 %

ड. गाँऊ/नगरपालिकाको हालको कर्मचारीहरू विवरण*

कर्मचारी	पालिका अन्तर्गत संख्या (महिला)	पालिका अन्तर्गत संख्या (पुरुष)	पुनःनिर्माण अन्तर्गत संख्या (महिला)	पुनःनिर्माण अन्तर्गत संख्या (पुरुष)
इन्जिनियर	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value
सब-इन्जिनियर	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value
ASSISTANT सब-इन्जिनियर	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value
अमीन	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value
सामाजिक परिचालक	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value
नगरपालिकाको कुल कर्मचारी	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value	* Please enter "0" if there is no value

च. गाँऊ/नगरपालिका क्षेत्रमा उपलब्ध प्राविधिक संख्या (अनुमानित)

प्राविधिक	संख्या (महिला)	संख्या (पुरुष)
घर निर्माणमा संलग्न डकर्मी / ठेकेदारहरू	* Please enter "0" if there is no value	* Please enter "0" if there is no value

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3/15

7/1/2020

Municipality Baseline Information Collection Checklist

परामर्शदाता इन्जिनियर (CONSULTANT ENGINEER)	*	*
Please enter "0" if there is no value	Please enter "0" if there is no value	
परामर्शदाता संस्था (CONSULTING FIRMS)		
Please enter "0" if there is no value		
छ. उपलब्ध तालिम प्राप्त जनशक्ति		
के गाउँ / नगरपालिकाका इन्जिनियरहरु भवन निर्माण संहिता कार्यान्वयन तथा नक्शापास सम्बन्धि तालिम प्राप्त छन?		
☐ छ ☐ छैन		
कति छन?		
Please enter "0" if there is no value		
के गाउँ / नगरपालिकाका सामाजिक परिचालाकहरु भवन निर्माण संहिता कार्यान्वयन तथा नक्शापास सम्बन्धि तालिम प्राप्त/ सचेत छन?		
☐ छ ☐ छैन		
कति छन?		
Please enter "0" if there is no value		
के गाउँ / नगरपालिकाका अन्य कर्मचारीहरु भवन निर्माण संहिता कार्यान्वयन तथा नक्शापास सम्बन्धि सचेत छन?		
☐ पूर्ण रुपमा सचेत छन ☐ केहि सचेत छन ☐ कतिपानि सचेत छैनन्		
तालिम	तालिम संख्या	महिला
		पुरुष
भुकम्प प्रतिरोधी निर्माणमा यस क्षेत्रका डकर्मी / ठेकेदारहरुलाई बिगतमा तालिम प्रदान गरिएको छ?		
☐ छ ☐ छैन		

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4/15

7/1/2020

Municipality Baseline Information Collection Checklist

तालिम संख्या	*
महिला	*
पुरुष	*
तालिम कस्ले संचालन गरेको ☐ गाँऊ / नगरपालिका आफैले ☐ संस्थाले सहयोग गरेको ☐ अन्य संस्थाले आयोजना गरेको	*
संस्थाको नाम	*
भुकम्प प्रतिरोधी भवन डिजाइन तथा निर्माणमा, CONSULTANT, DESIGNER, ENGINEER हरुलाई बिगतमा तालिम प्रदान गरिएको छ? ☐ छ ☐ छैन	*
तालिम संख्या	*
महिला	*
पुरुष	*
तालिम कस्ले संचालन गरेको ☐ गाँऊ / नगरपालिका आफैले ☐ संस्थाले सहयोग गरेको ☐ अन्य संस्थाले आयोजना गरेको	*

<https://enketo.ona.io/x/#pxQE0caV>

5/15

7/1/2020

Municipality Baseline Information Collection Checklist

संस्थाको नाम	*
के गाउँ / नगरपालिकाका क्षेत्रका इन्जिनियरहरु घर प्रबलिकरण प्रविधिमा तालिम प्राप्त छन?	*
☐ छ ☐ छैन	
तालिम संख्या	*
महिला	*
पुरुष	*
तालिम कस्ले संचालन गरेको	*
☐ गाँऊ / नगरपालिका आफैले ☐ संस्थाले सहयोग गरेको ☐ अन्य संस्थाले आयोजना गरेको	
संस्थाको नाम	*
के गाउँ / नगरपालिकाका क्षेत्रका डकर्मीहरु घर प्रबलिकरण प्रविधिमा तालिम प्राप्त छन?	*
☐ छ ☐ छैन	
तालिम संख्या	*
महिला	*

<https://enketo.ona.io/x/#pxQE0caV>

6/15

7/1/2020

Municipality Baseline Information Collection Checklist

पुरुष	*
तालिम कस्ले संचालन गरेको ☐ गाँऊ / नगरपालिका आफैले ☐ संस्थाले सहयोग गरेको ☐ अन्य संस्थाले आयोजना गरेको	*
संस्थाको नाम	*
भुकम्प प्रतिरोधी निर्माणमा घरधनीहरुलाई बिगतमा कुनै अभिमुखीकरण वा तालिम प्रदान गरेको छ? ☐ छ ☐ छैन	*
गाउँ / नगरपालिकाले भवन निर्माण स्थलमा नै गई सुरक्षित भवन निर्माण सम्बन्धि प्राविधिक सल्लाह दिने कार्यक्रम गरेको छ? ☐ छ ☐ छैन	*
गाउँ / नगरपालिका क्षेत्रमा सामजिक अगुवा तथा प्रबुद्ध बर्गका लागि सुरक्षित भवन निर्माण तथा नक्सा पास सम्बन्धि अभिमुखीकरण वा तालिमहरु भएको छ? ☐ छ ☐ छैन	*
निर्वाचित पदाधिकारीहरु तथा अधिकृतहरुलाई विपद व्यवस्थापन तथा सहरी योजना सम्बन्धी अभिमुखीकरण वा तालिमहरु भएको छ? ☐ छ ☐ छैन	*
के तपाईंको गाउँ / नगरपालिकामा भुकम्प प्रतिरोधी निर्माणमा ईन्जिनियर, डकर्मी तथा ठेकेदारहरुलाई तालिम प्रदान गर्ने कुनै निजी संघसंस्था (PRIVATE INSTITUTIONS) छ? ☐ छ ☐ छैन	*

<https://enketo.ona.io/x/#pxQE0caV>

7/15

7/1/2020

Municipality Baseline Information Collection Checklist

निजि संघसंस्था (PRIVATE INSTITUTIONS) को नाम उल्लेख गर्नुहोस।	*
के तपाईंको गाउँ / नगरपालिकामा भुकम्प प्रतिरोधी निर्माणमा ईन्जिनियर, डकर्मी तथा ठेकेदारहरुलाई तालिम प्रदान गर्ने कुनै सामाजिक तथा गैर सरकारी संस्थाहरु (NGOS, CBOS) छन्?	*
☐ छ ☐ छैन	
सामाजिक तथा गैर सरकारी संस्थाहरु (NGOS, CBOS) को नाम उल्लेख गर्नुहोस।	*
यस गाउँ / नगरपालिकामा बिगतमा सुरक्षित भवन निर्माण प्रवर्द्धनका लागि तल उल्लेखित अन्तरक्रिया छलफल र कार्यशाला गोष्ठीहरु भएका छन्? भएकोमा ठिक लगाउनुहोस।	*
☐ व्यवस्थित नक्सा पास सम्बन्धी ☐ सुरक्षित भवनका लागि कर्जा सम्बन्धी ☐ भवन निर्माण संहिता परिमार्जनका लागि ☐ जोखिम संवेदनशील भूउपयोग योजना (RSLUP) ☐ स्थानिय स्तरको विपद जोखिम व्यवस्थापन योजना (LDRMP/LDCRP) ☐ अन्य	
अन्य भए उल्लेख गर्नुहोस।	*
यस गाउँ / नगरपालिका क्षेत्रमा जोखिमको आधारमा आवास कर्जा (RISK BASED HOUSING LOAN / RISK INSURANCE SYSTEM) को व्यवस्था छ?	*
☐ छ ☐ छैन	
ज. भवन निर्माण संहिता कार्यान्वयन तथा नक्सा पासका लागि संस्थागत प्रणाली	
तपाईंको गाउँ / नगरपालिकामा भवन नियमावाली (BY-LAWS) छ?	*
☐ छ ☐ छैन	

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8/15

7/1/2020

Municipality Baseline Information Collection Checklist

कहिले बनाएको हो? <i>Please enter the date in BS</i>	*
तपाईंको गाउँ / नगरपालिकामा जोखिम संवेदनशील भूउपयोग योजना (RSLUP) बनेको छ? ☐ छ ☐ छैन	*
एकिकृत शहरी बिकाश योजना बनेको छ? ☐ छ ☐ छैन	*
बिपद व्यवस्थापन योजना तथा बिपद व्यवस्थापन कार्य योजना (DRM/ DM ACTION PLAN) बनेको छ? ☐ छ ☐ छैन	*
तपाईंको गाउँ / नगरपालिकामा पुराना नक्सा पास नभई निर्माण भएका वा अन्य प्रयोजनको लागि भवनहरुको अभिलेखिकरण प्रक्रिया गरिएको छ? ☐ छ ☐ छैन ☐ सुरु गर्ने योजना छ	*
गाउँ / नगरपालिकामा भूउपग्रह तस्बिर तथा जी.आई.एस (GIS) नक्सा कुन कुन छन? ☐ भूउपग्रह तस्बिर (Satellite Image) ☐ जी.आई.एस नक्सा (GIS Image)	*
GIS IMAGE मा कस्तो विवरण छ? ☐ भवन फुट प्रिन्ट्स ☐ संरचना मात्र	*

झ. भवन निर्माण संहिता कार्यान्वयन तथा नक्सा पासका प्रणालीको जानकारी

<https://enketo.ona.io/x/#pxQE0caV>

9/15

7/1/2020

Municipality Baseline Information Collection Checklist

भवन निर्माण संहिता कार्यान्वयनको अवस्था	*
☐ Declared and implemented ☐ Declared but not implemented ☐ Planning to declare ☐ Not Declared / No Interest	
भवन निर्माण संहिता कहाँ कहाँ कार्यान्वयन गरिएको छ?	*
☐ शहरी क्षेत्रमा ☐ शहर उन्मुख क्षेत्रमा ☐ ग्रामिण क्षेत्रमा ☐ सबै क्षेत्रमा	
नक्सा पासको प्रक्रिया सबै वडाहरुमा लागु छ?	*
☐ छ ☐ छैन	
नक्सा पासको प्रक्रिया लागु नभएका वडाहरु ।	*

के गाउँ / नगरपालिकामा स्थानिय जोखिम व्यवस्थापन योजना (LDRMP) छ?	*
☐ छ ☐ छैन	
गाउँ / नगरपालिकाको हालको नक्सा पास प्रणाली के छ?	*
☐ Building Permit Unit/Cell ☐ Separate Building Code Implementation (BCI) Cell ☐ Separate Earthquake Safety Unit ☐ Dedicated Technical Staff for BCI ☐ None ☐ Other	
अन्य भए उल्लेख गर्नुहोस।	*

<https://enketo.ona.io/x/#pxQE0caV>

10/15

7/1/2020

Municipality Baseline Information Collection Checklist

नक्सा पास प्रक्रिया सम्पन्न गर्न अनुमानित कति समय लाग्छ ? ☐ दुई हप्ता भित्र ☐ दुई - चार हप्ता ☐ पाँच - आठ हप्ता ☐ दुई महिना भन्दा बढी ☐ भन्न सकिन्न	*
के गाउँ / नगरपालिकामा भवन निर्माण संहिता कार्यान्वयनका लागि सल्लाहकार समिति वा प्राविधिक सल्लाहकार समिति छ? ☐ छ ☐ छैन	*
के गाउँ / नगरपालिकासँग भुकम्पिय क्षति परिदृष्य तथा कार्ययोजना (EARTHQUAKE DAMAGE SCENARIO AND ACTION PLANNING) छ? ☐ छ ☐ छैन	*
के नक्सा पास प्रक्रियामा "सी" श्रेणी को भवन निर्माणको लागि STRUCTURAL DRAWING र "ए" र "बि" श्रेणी को भवन निर्माणको लागि बिस्तृत डिजाइन आवश्यक पर्छ? (DO BUILDING PERMIT PROCESS NEED STRUCTURAL DRAWING FOR BUILDING CLASS "C" AND DETAIL DESIGNS FOR CLASS "A" AND "B" BUILDINGS?) ☐ छ ☐ छैन	*
के "बि" श्रेणीका भवन डिजाइन गर्नको लागि गाउँ / नगरपालिका क्षेत्रका प्राविधिकहरु सक्षम छन् ? ☐ छ ☐ छैन	*
छन भने, कति जना छन?	*

<https://enketo.ona.io/x/#pxQE0caV>

11/15

7/1/2020

Municipality Baseline Information Collection Checklist

के गाउँ / नगरपालिकाले भवन संरचना कम्प्लायन्स परिक्षण गर्न DETAILED चेकलिष्ट प्रयोग गर्छ? (DOES THE MUNICIPALITY USE DETAIL CHECKLIST FOR BUILDING CODE COMPLIANCE CHECK (BLD. CONFIGURATION, BLD. STRENGTH, BUILDING DUCTILE DETAILING) ? ☐ छ ☐ छैन	*
के संरचनागत कुराहरुलाई फिल्डमा गएर परिक्षण गर्ने गाउँ / नगरपालिकाको व्यवस्था छ? (PROVISION OF FIELD INSPECTION FOR STRUCTURAL DETAILS) ? ☐ छ ☐ छैन	*
कुन तहमा? ☐ Foundation ☐ Plinth Level ☐ Each Storey Structure ☐ Final Completion	*
के गाउँ / नगरपालिकासँग तालिम प्राप्त डकर्मीहरुको दर्ता गर्ने र सुची बनाउने व्यवस्था छ? ☐ छ ☐ छैन	*
के गाउँ / नगरपालिकासँग हालका कमजोर भवनहरुलाई बलियो बनाउन वा प्रबलिकरण गर्ने व्यवस्था छ? (WHILE ADDITION OF STOREY OR IN ANY CASE)? ☐ छ ☐ छैन	*
अन्दाजी कति प्रतिशत नक्साहरुभवन निर्माण संहिता अनुरूप पेश हुन्छन्? (SUBMITTED % OF DRAWINGS COMPLYING WITH BUILDING CODE) ? प्रतिशतमा(%) Please enter "0" if there is no value.	*
भवन निर्माण संहिता अनुरूप पेश भएका मध्ये अन्दाजी कति प्रतिशत नक्साहरुभवन निर्माण संहिता अनुरूप पास हुन्छन्? (APPROVED % OF DRAWINGS COMPLYING BUILDING CODE) ? प्रतिशतमा(%) Please enter "0" if there is no value.	*

<https://enketo.ona.io/x/#pxQE0caV>

12/15

7/1/2020

Municipality Baseline Information Collection Checklist

नक्सा पासमा आएका कति प्रतिशत घरहरु निर्माण संहिता अनुरूप बन्छन्? (FIELD LEVEL COMPLIANCE) ? प्रतिशतमा(%) Please enter "0" if there is no value.	*
यस गाउँ / नगरपालिकामा सुरक्षित निर्माणको प्रवर्द्धन र नियमको लागि COMPUTER SOFTWARE प्रणाली वा MOBILE APPLICATION को प्रयोग गर्न थालिएको छ कि छैन? ☐ छ ☐ छैन	*
गाउँ / नगरपालिकाले भवन निर्माण संहिता कार्यन्वयनको व्यवस्थामा केहि योजना बनाएको छ? ☐ छ ☐ छैन	*
बनाएको छन भने: के के बनाएको छ, ठिक चिन्ह लगाउनुहोस। ☐ Mason Training ☐ Engineers Training ☐ Awareness Program ☐ Demonstration Program ☐ Radio/TV Program ☐ Interaction Program ☐ Workshops/ Review of BCI ☐ Others	*
अन्य भए उल्लेख गर्नुहो।	*
गाउँ / नगरपालिकाले घर निर्माणको नक्सा बनाउने प्राविधिकहरुको कुनै खास योग्यता तोकेको छ? ☐ छ ☐ छैन	*
छ भने। कुन हो? ☐ Engineer ☐ Sub-Engineer	*

<https://enketo.ona.io/x/#pxQE0caV>

13/15

7/1/2020

Municipality Baseline Information Collection Checklist

के यस गाउँ / नगरपालिकामा निर्माणसंग सम्बन्धीत आर्किटेक्ट, ईन्जिनियर, योजनाबिद्, सुपरिवेक्षक सुचिकृत गर्न संस्थागत प्रणाली छ? (IS THERE AN ORGANIZED SYSTEM OF REGISTRATION OF CONSTRUCTION PRACTITIONERS, E.G., ARCHITECTS, ENGINEERS, TOWN PLANNERS, AND CONSTRUCTION SUPERVISORS IN YOUR MUNICIPALITY?) ☐ छ ☐ छैन	*
यस नगरपालिकामा भवन निर्माण संहिताको कार्यान्वयनका गर्ने वातावरण कस्तो छ? ☐ राम्रो छ ☐ ठीकै छ ☐ राम्रो छैन	*
राजनितिक समुह वा समाजिक संस्थाको सहयोग र चाहना छ कि छैन? ☐ छ ☐ छैन	*
गाउँ / नगरपालिकाको बजेट (रु)	*
नक्सापासबाट आयस्रोत (रु)	*
भवन निर्माण संहिता कार्यान्वयनका लागि बजेट बिनियोजन (रु)	
ज. संजाल तथा माध्यम	
यस गाउँ / नगरपालिका क्षेत्रमा भवन निर्माण संहिता कार्यान्वयनमा कुनै संस्थाले काम गरिरहेका छन्? ☐ छ ☐ छैन	*
कति वटा संस्था छन्?	*

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14/15

7/1/2020

Municipality Baseline Information Collection Checklist

संस्थाको नामहरु *
भवन निर्माण संहिता कार्यान्वयन तथा सुरक्षित भवन निर्माण प्रवर्द्धनका लागि निम्न उल्लेखित कार्यहरु गर्ने गरिएको छ? * ☐ अनुभवहरुको आदानप्रदानका लागि नगरपालिकाहरुको बिचमा अध्ययन भ्रमण ☐ अनुभवहरुको आदानप्रदान कार्यशाला ☐ भवन निर्माण संहिता कार्यान्वयन कार्यरत संस्थाहरु बीच नियमित बैठक ☐ छलफल तथा अन्तर्क्रिया कार्यक्रम ☐ अन्य
अन्य भए उल्लेख गर्नुहोस। *
THANK YOU FOR YOUR PARTICIPATION. THAT IS THE END OF THE QUESTIONNAIRE. WE WILL SHARE THE FINDINGS OF THIS STUDY THROUGH OUR WEBSITE WWW.HRRPNEPAL.ORG AS WELL AS THROUGH THE EMAIL ADDRESS IF PROVIDED ABOVE AND WOULD ENCOURAGE YOU TO SUBSCRIBE TO OUR COMMUNICATIONS FOR FUTURE UPDATES.
अन्तर्वाता लिने व्यक्तिको लागि यदि अन्तर्वाताको बारेमा केहि छ भने, उल्लेख गर्नुहोस।
फारम भर्ने व्यक्तिको नाम: * <i>Please include your agency name in parenthesis () if other than HRRP.</i>

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15/15



Report by:



Primarily funded by:



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